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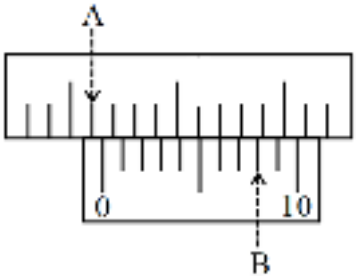
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PHYSICS

1) A vernier calipers used for measurement has it 9 main scale divisions equivalent to 10 vernier scale division and 1 cm on main scale is divided into 10 equal divisions. Figure shows the position of two scales at some instant. The distance between divisions marked with A and B is given by



(1) 7.3 mm

(2) 7.7 mm

(3) 8.1 mm

(4) 8.5 mm

2) For a wire, the Young's modulus(Y) is given by an equation $Y = (5 \times 10^4)\frac{M}{\ell}\frac{dyne}{cm^2}$, where M and □ are mass of wire and extension of wire respectively. Error in young's modulus is estimated by taking data from M–□ plot in graph paper. The smallest division on load axis and extension axis are 20g and 0.08cm respectively. If mass used is 500g and □ is 4 cm then percentage error in Y is :

(1) 2.5%

(2) 3%

(3) 4%

(4) 6%

3) In an experiment to determine coefficient of viscosity (η) of liquid using a spherical ball of radius R consider the following statements:
(A) The terminal velocities of balls of different diameter but identical material is constant for a given liquid.
(B) Terminal velocity decreases if temperature of liquid is raised.
(C) If the balls are dropped with some initial speed in liquid, the value of η will change.
(D) Momentum of ball at terminal velocity is proportional to R^5.
Which of the following sets are true ?

(1) A, B & D only

(2) A, C and D only

(3) B, C and D only

(4) D only

4) Two liquids A and B are at temperature of 84°C and 24°C. Their masses are in the ratio of 3 : 5

while their specific heats are in the ratio 4 : 3. If these liquids are mixed then resulting temperature of mixture is :

(1) 40°C

(2) 50°C

(3) 45°C

(4) 60°C

5) A student connects voltmeter and ammeter as shown in circuit figure (a) to measure the resistance R of a wire. The V-I characteristics thus obtained is shown in figure (b). The value of resistance (R) is:

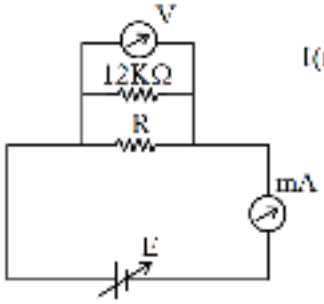


Fig.(a)

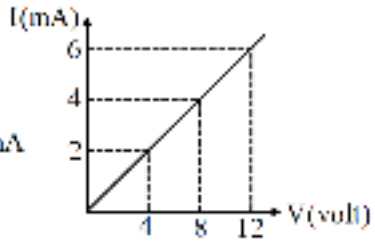


Fig.(b)

(1) 1600 Ω

(2) 2000 Ω

(3) 2400 Ω

(4) 2600 Ω

6) A light ray passes through a prism under condition of minimum deviation. Consider the following statements:
(A) The angle of incidence becomes equal to angle of emergence.
(B) There are always two sets of angle of incidence for which deviation will be same except under condition of minimum deviation.
(C) Larger angle prisms provide smaller angle of minimum deviation.
(D) Angle of refraction becomes twice the angle of prism.
Choose the set of correct statements:

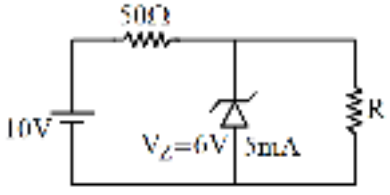
(1) A, B and C

(2) A, B and D

(3) A, C and D

(4) A and B only

7) In the given circuit, a 6V zener diode has negligible resistance for a knee current 5mA. The minimum value of R (in Ω) so that voltage across it, does not fall below 6V is:



(1) 40

(2) 60

(3) 75

(4) 80

8) An electric dipole with dipole moment $\vec{P} = (2\hat{i} + 3\hat{j})$C-m is kept in an electric field of $\vec{E} = 4\hat{i}$N/C. The torque acting on it is:

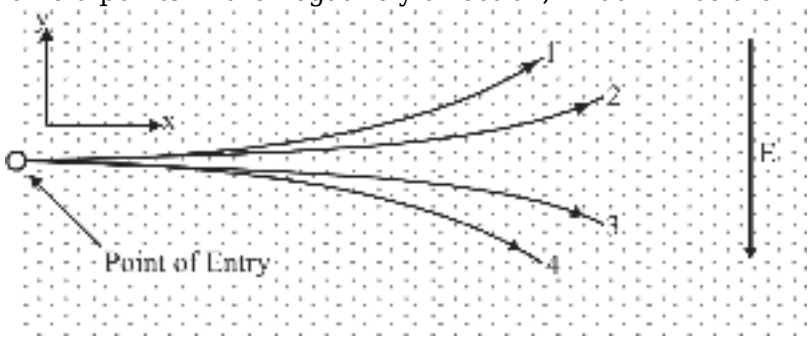
(1) $-12\hat{k}$(Nm)

(2) $8\hat{k}$ (Nm)

(3) $12\hat{k}$(Nm)

(4) $-8\hat{k}$ (Nm)

9) Two particles of the same mass carry charges +3Q and –2Q respectively. They are shot into a region that contains a uniform electric field one after the other as shown. The particles have the same initial velocities in the positive x direction. The lines, numbered 1 through 4, indicate possible paths for the particles. If the electric field points in the negative y direction, what will be the resulting paths for these particles?



(1) path 1 for +3Q and path 3 for –2Q

(2) path 3 for +3Q and path 3 for –2Q

(3) path 2 for +3Q and path 4 for –2Q

(4) path 4 for +3Q and path 2 for –2Q

10) Two equal charges q are placed at a distance of '2a' and a third charge –2q is placed at the midpoint. The potential energy of the system is :-

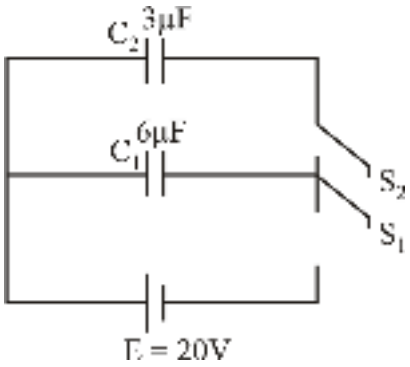
(1) $\frac{q^2}{8\pi\varepsilon_0 a}$

(2) $\frac{6q^2}{8\pi\varepsilon_0 a}$

(3) $-\frac{7q^2}{8\pi\varepsilon_0 a}$

(4) $-\frac{9q^2}{8\pi\varepsilon_0 a}$

11) In the circuit shown in Fig. C_1 = 6 µF, C_2=3 µF and battery E = 20 V. The switch S_1 is first closed. It is then opened and afterwards S_2 is closed. What is the final charge on C_2 :-



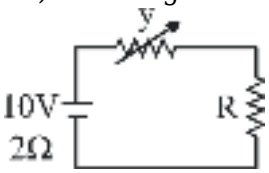
(1) 120 μC

(2) 40 μC

(3) 80 μC

(4) 20 μC

12) In the figure shown the power generated in y is maximum when y = 5Ω. Then R is



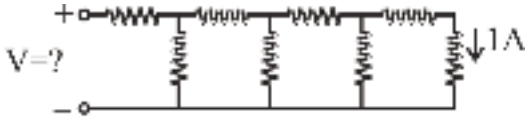
(1) 2 Ω

(2) 6 Ω

(3) 5 Ω

(4) 3 Ω

13) Each element in the finite chain of resistors shown in the figure is 1Ω. A current of 1 A flows through the final element. Then what is the potential difference V across input terminals of the chain.



(1) 12 volt

(2) 34 volt

(3) 1 volt

(4) 16 volt

14) Consider a wire having current 10A having area of cross-section $1cm^2$. If number of electrons per unit volume is $9 \times 10^{28} m^{-3}$. Find the drift velocity of electrons :-

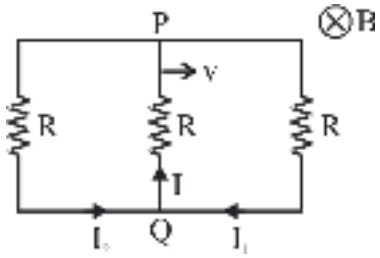
(1) 6.94×10^{-6} m/s

(2) 1.94×10^{-3} m/s

(3) 5.94×10^{-2} m/s

(4) 2.94×10^{-4} m/s

15) A rectangular loop has a sliding connector PQ of length and resistance R and is moving with a speed v as shown in the figure. The set up is placed in a uniform magnetic field going into the plane



of the paper. The three currents are :-

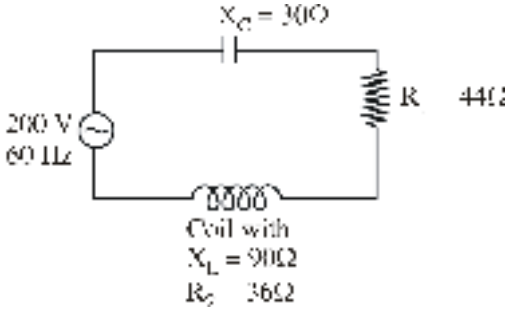
(1) $I_1 = I_2 = \frac{B\ell v}{6R};\ I = \frac{B\ell V}{3R}$

(2) $I_1 = -I_2 = \frac{B\ell v}{R};\ I = \frac{2B\ell V}{R}$

(3) $I_1 = I_2 = \frac{B\ell v}{3R};\ I = \frac{2B\ell V}{3R}$

(4) $I_1 = I_2 = I = \frac{B\ell V}{R}$

16) As given in the figure, a series circuit connected across a 200 V, 60 Hz line consists of a capacitor of capacitive reactance 30 Ω, a non-inductive resistor of 44 Ω, and a coil of inductive reactance 90Ω and resistance 36Ω. The power dissipated in the coil is



(1) 320 W

(2) 176 W

(3) 144 W

(4) 0 W

17) The electric field of a plane electromagnetic wave varies with time of amplitude 2 V/m vibrating along z-axis. The average energy density of magnetic field is (in J/m^3) :

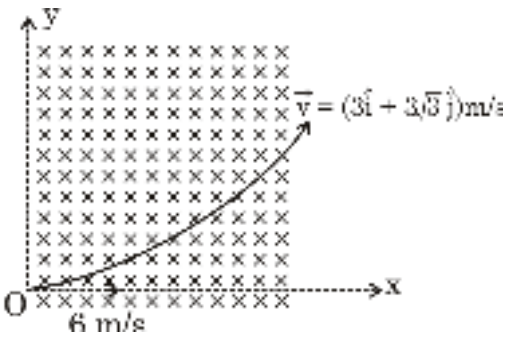
(1) 13.29×10^{-12}

(2) 8.85×10^{-12}

(3) 17.72×10^{-12}

(4) 4.43×10^{-12}

18) A particle of charge 'q' and mass 'm' enters a uniform magnetic field region $\vec{B} = -B\hat{k}$ at origin 'O' with a velocity $\vec{v}_0 = 6\hat{i}$ m/s and after some time it exits the magnetic field region with a velocity $\vec{v} = \left(3\hat{i} + 3\sqrt{3}\hat{j}\right)$ m/s as shown. The time interval for which the particle has moved in the magnetic

field region is :- 

(1) $\frac{\pi m}{qB}$

(2) $\frac{\pi m}{2qB}$

(3) $\frac{\pi m}{3qB}$

(4) $\frac{\pi m}{4qB}$

19) A circular current carrying coil has a radius R. The distance from the centre of the coil on the axis where the magnetic induction will be $\frac{1}{8}$th of its value at the centre of the coil, is

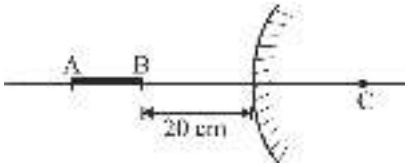
(1) $\frac{R}{\sqrt{3}}$

(2) $R\sqrt{3}$

(3) $2\sqrt{3}R$

(4) $\frac{2}{\sqrt{3}}R$

20) A convex mirror of focal length 10 cm is shown in figure. A linear object AB = 5 cm is placed along the optical axis. Point B is at distance 20 cm from the pole of mirror. Then length of image of AB will be :-



(1) $\frac{5}{14}$cm

(2) $\frac{10}{21}$cm

(3) $\frac{10}{14}$cm

(4) 2.5 cm

21) Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : For a simple microscope, the angular size of the object equals the angular size of the image.

Reason R : Magnification is achieved as the small object can be kept much closer than 25 cm to the

eye and hence it subtends a large angle.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) A is true but R is false

(2) Both A and R are true but R is NOT the correct explanation of A.

(3) Both A and R are true and R is the correct explanation of A

(4) A is false but R is true

22) In Young's double slit experiment the two slits are d distance apart. Interference pattern is observed on a screen at a distance $D >> d$ from the slits. A dark fringe is observed on the screen directly opposite to one of the slits. The wavelength of light is :-

(1) $\frac{D^2}{2d}$

(2) $\frac{d^2}{2D}$

(3) $\frac{D^2}{d}$

(4) $\frac{d^2}{D}$

23) When a metal is irradiated by monochromatic light, the maximum kinetic energy of the photo–electrons is 1.2eV. If frequency of the light is increased by 50% then maximum kinetic energy of photo–electron is 3.6eV. Evaluate the workfunction of the metal.:-

(1) 3.6 eV

(2) 5.8 eV

(3) 9.7 eV

(4) None of these

24) The radius of electron in a H-atom changes from 2.12 Å to 4.77 Å. The H-atom has :-

(1) Absorbed a photon of energy 1.51 eV

(2) Absorbed a photon of energy 1.89 eV

(3) Emitted a photon of energy 10.2 eV

(4) Emitted a photon of energy 1.04 eV

25) An electron moves along circular path of radius 'r' in uniform transverse magnetic field 'B' then de-Broglie wave length associated with it –

(1) $\frac{h}{Br}$

(2) $\frac{he}{Br}$

(3) $\frac{h}{eBr}$

(4) $\frac{hB}{er}$

26) How much nuclei of U^{235} is to fission to produce 1KJ energy. Let each fission produces 200 MeV energy

(1) 3.1×10^{10}

(2) 3.1×10^{13}

(3) 3.1×10^{16}

(4) 3.1×10^{20}

27) What is the power output of ${}_{92}U^{235}$ reactor if it taken 30 days to use up 2kg of fuel, and if each fission gives 185 MeV of usable energy :

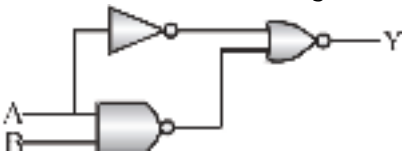
(1) 58.46 MW

(2) 40×10^4 W

(3) 38.64×10^3 W

(4) 62×10^8 W

28) Truth table for the given arrangement is :-



(1)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(2)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(3)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(4)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

29) A particle is moving on x-axis has potential energy U = 2 – 20 x + $5x^2$ joule. The particle is released at x = – 3. The maximum value of 'x' will be [x is in meter and U is in joule] :-

(1) 5 m
(2) 3 m
(3) 7 m
(4) 8 m

30) Two vibrating strings A and B of same material but lengths 2L and 3L have radii 3r and 2r respectively. They are stretched under same tension. String A vibrates in fundamental mode and string B in second overtone. The ratio of their frequencies n_A/n_B will be :

(1) 1 : 2
(2) 1 : 3
(3) 1 : 4
(4) 2 : 3

31) A diatomic ideal gas is heated at constant volume till pressure is doubled again heated at constant pressure till its volume is become doubled. The average molar heat capacity for whole process is:

(1) $\frac{13R}{6}$
(2) $\frac{19R}{6}$
(3) $\frac{23R}{6}$
(4) $\frac{17R}{6}$

32) A solid copper cube of edges 1 cm is suspended in an evacuated enclosure. Its temperature is found to fall from 100ºC to 99ºC in 100 s. Another solid copper cube of edges 2 cm, with similar

surface nature, is suspended in a similar manner. The time required for this cube to cool from 100ºC to 99ºC will be approximately -

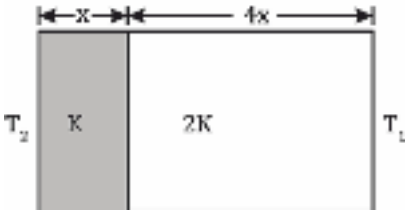
(1) 25 s

(2) 50 s

(3) 200 s

(4) 400 s

33) The temperature of the two outer surfaces of a composite slab consisting of two materials having coefficients of thermal conductivity K and 2K and thickness x and 4x respectively are T_2 and T_1 ($T_2 > T_1$) The rate of heat transfer through the slab in steady state is $\frac{AK(T_2 - T_1)}{x}_f$. Where f is equal to :



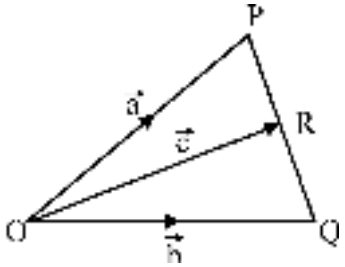
(1) 1

(2) 1/2

(3) 2/3

(4) 1/3

34) Figure shows three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ where R is the mid point of PQ. Which of the following relation is correct :-



(1) $\vec{a} + \vec{b} = 2\vec{c}$

(2) $\vec{a} + \vec{b} = \vec{c}$

(3) $\vec{a} - \vec{b} = 2\vec{c}$

(4) $\vec{a} - \vec{b} = \vec{c}$

35) A wave is propagating along x-axis in a medium, so that displacement y of particle at any position x, at any time t is given by : $y = \frac{a}{t^2} e^{-\left[\frac{t}{b} - \frac{x}{c}\right]}$, where a, b and c are constants.
Choose the correct statement :-

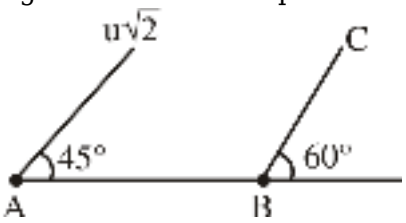
(1) dimension of b is [L]

(2) dimension of a is [LT^2]

(3) dimension of c is [T]

(4) $e^{-\left(\frac{t}{b} - \frac{x}{c}\right)}$ is not dimensionless

36) A particle is projected from point A with velocity $u\sqrt{2}$ at angle 45° to horizontal as shown in figure. It strikes the plane BC at right angle. What is the velocity of particle at the time of impact ?



(1) $\frac{\sqrt{3}u}{2}$

(2) $\frac{u}{2}$

(3) $\frac{2u}{\sqrt{3}}$

(4) u

37) From a tower of height H, a particle is thrown up with speed 5 m/s. The time taken to hit the ground is 3 times that of time to reach by it to highest point. Height of the tower H is. The value of H is

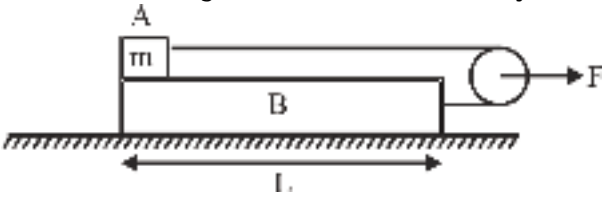
(1) H = 10.5 m

(2) H = 5.25 m

(3) H = 3.75 m

(4) H = 7.25 m

38) A small block of mass m is placed on a plank of mass 2m and length L. A force F is applied as shown in the figure. Find time taken by block A to reach end of plank B :-



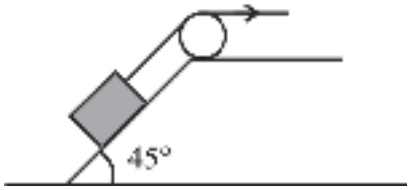
(1) $\sqrt{\frac{mL}{2F}}$

(2) $\sqrt{\frac{4mL}{F}}$

(3) $\sqrt{\frac{8mL}{F}}$

(4) $\sqrt{\frac{mL}{4F}}$

39) A block of mass 200 kg is being pulled up by men on an inclined plane at angle of 45° as shown. The coefficient of static friction is 0.5. Each man can only apply a maximum force of 500 N. Calculate the number of men required for the block to just start moving up the plane :-



(1) 10

(2) 15

(3) 5

(4) 3

40) A car of mass 1250 kg is moving at 30m/s. Its engine delivers 30 kW while resistive force due to surface is 750N. What max acceleration can be given in the car

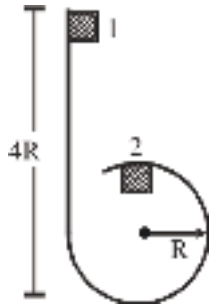
(1) $\frac{1}{3}\,m/s^2$

(2) $\frac{1}{4}m/s^2$

(3) $\frac{1}{5}\,m/s^2$

(4) $\frac{1}{6}\,m/s^2$

41) A cube of mass M starts at rest from point 1 at a height 4R, where R is the radius of the circular track. The cube slides down the frictionless track and around the loop. The force which the track exerts on the cube at point 2 is :



(1) 3 mg

(2) mg

(3) 2 mg

(4) cube will not reach the point 2.

42)

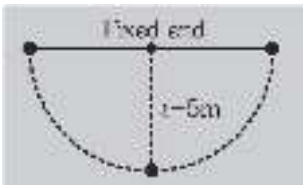
Two identical disks, with rotational inertia
(I =1/2 MR^2), rolls without sliding across a horizontal floor with the same speed and then climb up inclined planes of same inclination. Disk A rolls up its incline without sliding. On the other hand, disk B rolls up a frictionless incline. Otherwise the inclines are identical. Disk A reaches a height 12 cm above the floor before rolling down again. Disk B reaches a height above the floor of:

(1) 24 cm

(2) 18 cm

(3) 12 cm

(4) 8 cm

43) A mass of 4 kg is suspended from a steel wire of length 5 meter to form a pendulum arrangement. If the mass is moved to one side and released from the horizontal position of wire then find the maximum extension in the length of the wire. 

(Given $Y_{steel} = 2 \times 10^{11}$ N/m^2 and area of cross–section of wire = 2 mm^2, g = 10 m/s^2)

(1) 0.2 mm

(2) 0.8 mm

(3) 1.5 mm

(4) 0.9 mm

44) Near the earth's surface time period of a satellite is 1.4 hrs. Find its time period if it is at the distance '4R' from the centre of earth :-

(1) 32 hrs.

(2) $\left(\frac{1}{8\sqrt{2}}\right)$ hrs.

(3) $8\sqrt{2}$ hrs.

(4) 16 hrs.

45) A thin steel ring of inner radius r and cross-sectional area A is fitted on to a wooden disc of radius R (R > r). If Young's modulus be Y, then the tension in the steel ring is

(1) $AY\left(\frac{R}{r}\right)$

(2) $AY\left(\frac{R-r}{r}\right)$

(3) $\frac{Y}{A}\left(\frac{R-r}{r}\right)$

(4) $\frac{Yr}{AR}$

CHEMISTRY

1) Which of the following statement is incorrect?

(1) The energies of the various subshells in the same shell of H–atom are in order s < p < d < f

(2) s–orbital is non-directional

(3) Electrons behave as wave if they are in motion.

(4) A p–orbital can accomodate a maximum of 2 electrons with opposite spin.

2) PQ_2 dissociates as $PQ_2(g) \rightleftharpoons P(g) + 2Q(g)$
The initial pressure of PQ_2 is 600 mm Hg. At equilibrium the total pressure is 900 mm Hg. Calculate the partial pressure of Q at equilibrium in mm Hg :-

(1) 300
(2) 150
(3) 200
(4) 100

3) **Statement-1** : Rate of reaction is always positive.
Statement-2 : Negative sign in rate of disappearance of reactant represent rate of reaction decreases.

(1) Both Statement are correct
(2) Both Statement are incorrect
(3) Statement-1 is correct and Statement-2 is incorrect
(4) Statement-2 is correct and Statement-1 is incorrect.

4) When a certain conductivity cell was filled with KCl solution (conductivity = 3 $\Omega^{-1}m^{-1}$) it had a resistance of 100 Ω at 300 K temperature when filled with 0.01 M K_2SO_4 solution it has a resistance of 300Ω. Calculate molar conductivity of K_2SO_4 solution in terms of Ω^{-1} m^2mol^{-1}.

(1) 10^{-5}
(2) 10^{5}
(3) 10^{-1}
(4) 10^{-3}

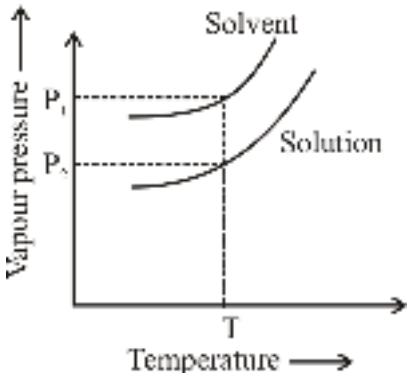
5) In the following which one has highest degree of hydrolysis.

(1) 0.01 M NH_4Cl
(2) 0.1 M NH_4Cl
(3) 0.001 M NH_4Cl
(4) Same

6) The osmotic pressure of equimolar solution of urea, $BaCl_2$ and $AlCl_3$ will be in the order :

(1) $AlCl_3 > BaCl_2 >$ urea
(2) $BaCl_2 > AlCl_3 >$ urea
(3) urea $> BaCl_2 > AlCl_3$
(4) $BaCl_2 >$ urea $> AlCl_3$

7) P_1 and P_2 are vapour pressure of solvent and solution at temperature T. The solution is prepared by dissolving non-volatile solute in the same solvent. P_1 = 760 torr, P_2 = 684 mm Hg, Mole fraction



of solvent in the solution will be:

(1) 0.1

(2) 0.2

(3) 0.9

(4) Can't be predicted

8) The volume of air (O_2 = 20%) necessary for the complete combustion of 20 L propane is :-

(1) 500 L

(2) 60 L

(3) 80 L

(4) 100 L

9) Select the correct match :

Column-I		**Column-II**	
(A)	22.4 L CH_4 at STP	(P)	Contains 8 N_A neutrons
(B)	18 g H_2O()	(Q)	Contains total 4 N_A atoms
(C)	17 g NH_3(g)	(R)	Contains 10 N_A electrons
(D)	10 g HF (atomic mass of F = 19)	(S)	Contains 0.5 mol molecule

(1) A → Q, B → S, C → R, D → P

(2) A → R, B → P, C → Q, D → S

(3) A → R, B → Q, C → P, D → S

(4) A → R, B → P, C → S, D → Q

10) What is the equivalent mass of IO_4^- when it is converted into I_2 in acidic medium?
(M = mol.wt of IO_4^-)

(1) M/6

(2) M/7

(3) M/5

(4) M/4

11) The number of moles of oxalate ions oxidised by one mole of $KMnO_4$ in acidic medium :-

(1) 5/2
(2) 2/5
(3) 3/5
(4) 5/3

12) The heat of combustion of ethylene $C_2H_4(g)$ is –1420 kJ/mole. The number of litres of C_2H_4 at NTP that would evolve 355 kJ on combustion is

(1) 2.8
(2) 8.4
(3) 5.6
(4) 11.2

13) One mole of ideal gas is allowed to expand reversibly and adiabatically from a temperature of 27°C. If the work done by the gas in the process is 3 kJ, the final temperature will be equal to (C_V=20 J/K mol)

(1) 100 K
(2) 450 K
(3) 150 K
(4) 400 K

14) Which of the following statements are correct :
(i) When freshly prepared precipitate are converted into colloidal solution then process is called peptization.
(ii) Lyophilic sol are mostly of inorganic nature.
(iii) Lyophilic sol are highly solvated as particles have great affinity for solvent.
(iv) Lyophobic sols are reversible sols so special methods are required to prepare them.

(1) Only i
(2) i and iii
(3) ii and iv
(4) All of these

15) The colour of phenolphthalein indicator in acid solution is

(1) Pink
(2) Yellow
(3) Colourless
(4) Orange

16) Give the correct initial T for true & F for false statement
(I) For Mn, the order of energy is
3s = 3p = 3d for shell number 3
(II) 32 elements are present in 5th period of long form periodic table

(III) For Zr^{+2} ion, $\mu_s = \sqrt{8}$ BM
(IV) General valence shell electronic configuration for *f*-block elements is
$(n-2)\,f^{1-14}(n-1)\,d^{10}\,ns^2$

(1) T T F T
(2) T F F T
(3) F F T F
(4) F T T F

17) Select the incorrect statements :

(1) Size of H^- is larger than F^-
(2) Rb is more electropositive as compared to Ca
(3) Na^+ is more electronegative than the Na
(4) Cl^- is more electronegative than that of F

18) Which of the following order is incorrect :-

(1) $SbH_3 > NH_3 > AsH_3 > PH_3$ Melting point
(2) HF > HI > HBr > HCl Boiling point
(3) $H_2O < H_2S < H_2Se$ (Acidic strength)
(4) $NH_3 > SbH_3 > AsH_3 > PH_3$ (Bond polarity)

19) Correct order of bond angle is :-
(a) $NH_4^+ > NH_2^-$ (b) $OF_2 < Cl_2O$
(c) $NO_2^+ < NO_2^-$ (d) $BF_3 < BCl_3$

(1) a, c, d
(2) a, b, c
(3) Only a, b
(4) Only c, d

20) The linear combination of atomic orbitals to form molecular orbitals takes place only when the combining atomic orbitals
A. have the same energy
B. have the minimum overlap
C. have same symmetry about the molecular axis
D. have different symmetry about the molecular axis
Choose the ***most appropriate*** from the options given below :

(1) A, B, C only
(2) A and C only
(3) B, C, D only
(4) B and D only

21) Match **List-I** with **List-II** :

List-I (Species)		List-II (Hybrid Orbitals)	
(a)	SF_4	(i)	sp^3d^2
(b)	$[Ni(NH_3)_6]^{2+}$	(ii)	d^2sp^3
(c)	NO_2^+	(iii)	sp^3d
(d)	MnO_4^-	(iv)	sp^3
		(v)	sp

Choose the **correct** answer from the options given below :

(1) (a)-(i), (b)-(ii), (c)-(v) and (d)-(iii)

(2) (a)-(iii), (b)-(ii), (c)-(v) and (d)-(iv)

(3) (a)-(iii), (b)-(i), (c)-(v) and (d)-(iv)

(4) (a)-(iv), (b)-(iii), (c)-(ii) and (d)-(v)

22) Select the correct statement :-

(1) $[Cr(gly)_3]$ can shows both Geometrical and optical isomerism

(2) $[Ni(CO)_4]$ have six bond angle of 109° 28'

(3) In $[Ni(CO)_4]$ both σ and π characters present in between metal and CO.

(4) All of these

23) **Statement–1 :** In complex $[Cr(NH_3)_4\ BrCl]Cl$, the 'spin only' magnetic moment is close to 1.73 B.M.

Statement–2 : All known mononuclear octahedral complexes of chromium(III), irrespective of the strength of the ligand field, must have three unpaired electrons.

(1) Statement- 1 is True, Statement- 2 is True; Statement- 2 is a correct explanation for Statement- 1.

(2) Statement- 1 is True, Statement- 2 is True; Statement- 2 is NOT a correct explanation for Statement- 1.

(3) Statement- 1 is True, Statement- 2 is False.

(4) Statement- 1 is False, Statement- 2 is True

24) Match List I with List II

LIST-I (Substances)		LIST-II (Element Present)	
(A)	Ziegler catalyst	(I)	Rhodium
(B)	Blood Pigment	(II)	Cobalt
(C)	Wilkinson catalyst	(III)	Iron
(D)	Vitamin B_{12}	(IV)	Titanium

Choose the correct answer from the options given below:

(1) A-II, B-IV, C-I, D-III

(2) A-II, B-III, C-IV, D-I

(3) A-III, B-II, C-IV, D-I

(4) A-IV, B-III, C-I, D-II

25) Which of the following statements are correct about Zn, Cd and Hg?
A. They exhibit high enthalpy of atomization as the d-subshell is full.
B. Zn and Cd do not show variable oxidation state while Hg shows +1 and + 2.
C. Compounds of Zn, Cd and Hg are paramagnetic in nature.
D. Zn, Cd and Hg are called soft metals.
Choose the ***most appropriate*** from the options given below:

(1) B, D only

(2) B, C only

(3) A, D only

(4) C, D only

26) Given below are two statements:
Statement I : CeO_2 can be used for oxidation of aldehydes and ketones.
Statement II : Aqueous solution of $EuSO_4$ is a strong reducing agent.
In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is false but statement II is true.

(2) Statment I is true but statement II is false.

(3) Both statement I and statement II are true.

(4) Both statement I and statement II are false.

27) Sometimes yellow turbidity appears while passing H_2S gas even in the absence of second group radicals. This is because :-

(1) Sulphur is present in the mixture as an impurity

(2) The fourth group radicals get precipitated as sulphides

(3) Oxidation of H_2S gas by some radicals acting as oxidant

(4) The third group radicals get precipitated as hydroxides

28) Graphite is good conductor of current but diamond is non-conductor because :

(1) Diamond is hard and graphite is soft

(2) Graphite and diamond have different atomic configuration

(3) Graphite is composed of positively charged carbon ions

(4) Graphite has hexagonal layer structure with mobile π-electrons while diamond has continuous tetrahedral covalent structure with no free electrons

29) Given below are two statements :
Statement I: Group 13 trivalent halides get easily hydrolyzed due to their covalent nature.
Statement II: $AlCl_3$ upon hydrolysis in acidified aqueous solution forms octahedral $[Al(H_2O)_6]^{3+}$ ion.

In the light of the above statements, choose the ***correct answer*** from the options given below :

(1) Statement I is true but statement II is false

(2) Statement I is false but statement II is true

(3) Both statement I and statement II are false

(4) Both statement I and statement II are true

30) Choose the correct match :-

	Column-I		**Column-II**
P	NaBr + conc. H_2SO_4	i	colourless gas
Q	Na_2S + dil HCl	ii	brown color
R	$NaNO_2$ + dil HCl	iii	rotten egg smell
S	$NaNO_3$ + conc. H_2SO_4	iv	paramagnetic

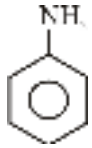
(1) P - ii ; Q - i, iii; R - ii, iv ; S - ii, iv

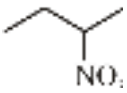
(2) P - ii ; Q - i, iii; R - ii, ; S - ii, iv

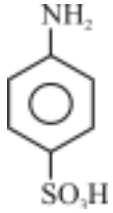
(3) P - ii ; Q - i, iii; R - ii, iv ; S - ii,

(4) P - ii ; Q - i, iii; R - i, iii ; S - ii, iv

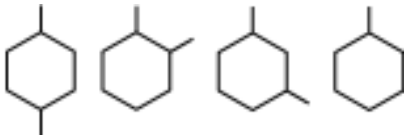
31) When sodium fusion extract of an organic compound treat with $FeSO_4$ solution and then heated with conc. H_2SO_4 & $FeCl_3$ it forms blood red but when Na mix in excess then it gives blue colour by same process, identity organic compound ?

(1) 

(2) 

(3) 

(4) 

32) Find out the total number of chiral centre in given compounds. 

(1) 7

(2) 6

(3) 5

(4) 4

33) Total number of sigma (σ) _____and pi(π)_____ bonds respectively present in hex-1-en-4-yne are :

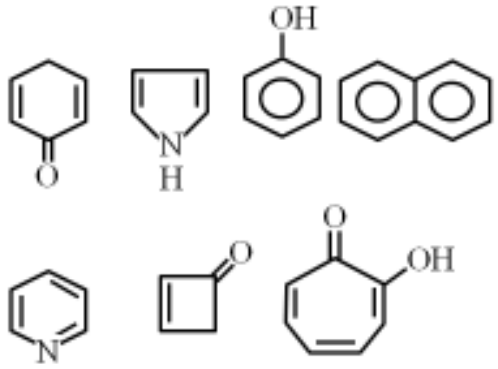
(1) 13 and 3

(2) 11 and 3

(3) 3 and 13

(4) 14 and 3

34) How many are aromatic compounds ?



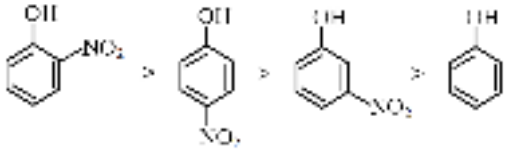
(1) 7

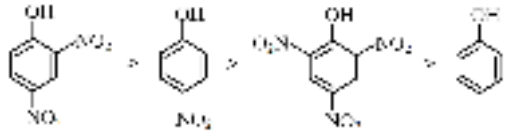
(2) 6

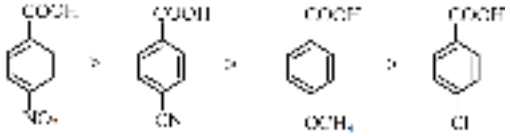
(3) 5

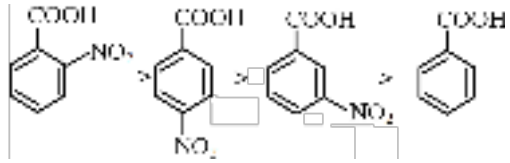
(4) 4

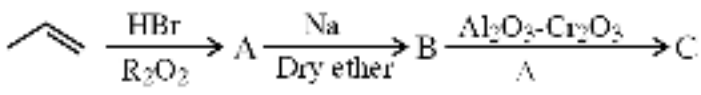
35) Which is correct order of acidic strength

(1) 

(2) 

(3) 

(4) 

36) 

C is :

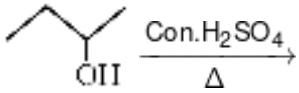
(1) n-Hexane

(2) Cyclohexane

(3) Benzene

(4) Toluene

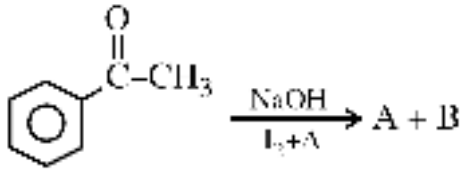
37) Which of the following reaction is not used to prepare an alkene ?

(1) 

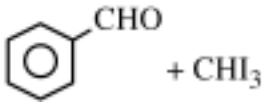
(2) $CH_3 - CH_2Cl \xrightarrow[\Delta]{alc.KOH}$

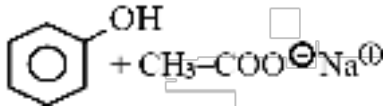
(3) $CH_3 - CH_2OH \xrightarrow[413K]{Con.H_2SO_4}$

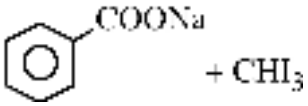
(4) $CH_3 - C \equiv C - CH_3 \xrightarrow{Na+Liq.NH_3}$

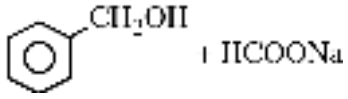
38) 

A and B will be :-

(1) 

(2) 

(3) 

(4) 

39) **Statement-I :** Sec. butyl chloride is more reactive than n-butyl chloride for S_{N^2} reaction.
Statement-II : Benzyl chloride is more reactive than ethyl chloride for S_{N^1} reaction.

(1) Both statement I and II are incorrect

(2) Both statement I and II are correct.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

40) Reactivities order of nucleophillic substitution reaction in acid derivatives compound is.

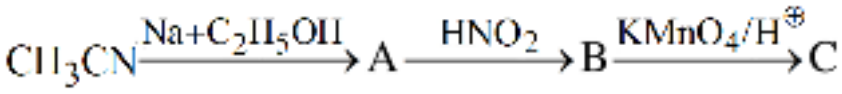
(1) Acid anhydride >Amide > Ester >Acyl chloride

(2) Acyl chloride >Ester>Acid anhydride > Amide

(3) Acyl chloride >Acid anhydride >Ester> Amide

(4) Ester >Acyl chloride>Amide >Acid anhydride

41) Identify product 'C' in the series -

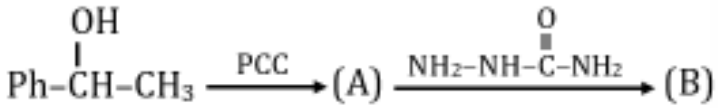


(1) CH_3COOH

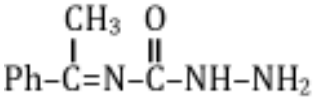
(2) $CH_3CH_2NH–OH$

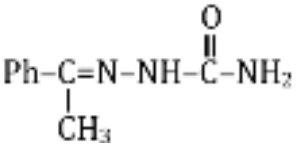
(3) CH_3CONH_2

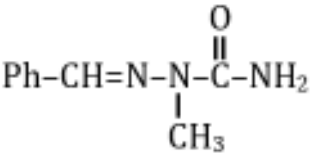
(4) CH_3CHO

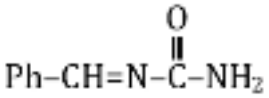
42) 

Product (B) is :-

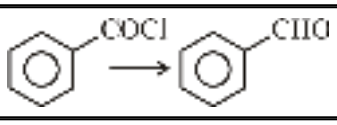
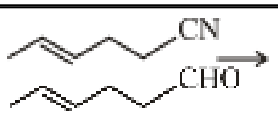
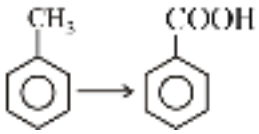
(1) 

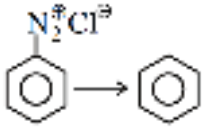
(2) 

(3) 

(4) 

43) Match the following :-

	Column-I (Reaction)		**Column-II (Reagent)**
(a)		(i)	Et–OH
(b)		(ii)	Hot $KMnO_4/H^+$
(c)		(iii)	DIBAL-H/ H_2O

(d)		(iv)	H_2, Pd-$BaSO_4$

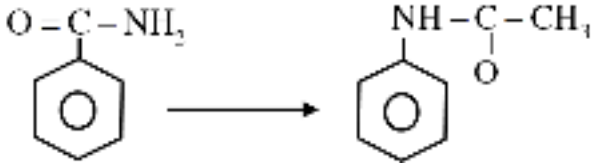
(1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(2) (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

(3) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(4) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

44) The reagents needed to convert



(1) (i) KOH, Br_2 ; (ii) $LiAlH_4$

(2) (i) KOH, Br_2 ; (ii) CH_3–COCl

(3) (i) HONO, Cu_2Cl_2, (ii) $(CH_3CO)_2O$

(4) (i) KOH, Br_2 ; (ii) Ni, H_2, (iii) CH_3COCl

45) Which vitamin is responsible for rickets

(1) Vitamin H

(2) Vitamin D

(3) Vitamin B-6

(4) Vitamin 6

BIOLOGY

1) In which of the following system of classification each character (vegetative and sexual) is given equal importance ?
(a) Artificial system
(b) Numerical system
(c) Natural system
(d) Phylogenetic system

(1) a and d

(2) only b

(3) a, b and c

(4) a and b

2)

Identify the correct match from the Column-I, II and III :-

Column-I	Column-II	Column-III

1	Biodiversity	a	Used in natural classification	i	Study of both external and internal characters
2	Gross morphology	b	Identification, nomenclature and classification	ii	Different type of plants and animals
3	Systematics	c	Includes phytochemistry	iii	Taxonomy
4	Neosystematics	d	Different types and number of species	iv	Includes ecological informations

(1) 1-d,ii 2-a,i 3-b,iii 4-c,iv

(2) 1-a,i 2-b,ii 3-c,iii 4-d,iv

(3) 1-d,iv 2-c,iii 3-b,ii 4-c,i

(4) 1-d,iii 2-c,ii 3-a, iv 4-b, i

3) Given below are two statements :-
Statement I :- Mycoplasma are the smallest living cells known and can survive without oxygen
Statement II :- Many Mycoplasma are pathogenic in animals and plants.
In the light of the above statements, Choose the most appropriate answer from the option given below :-

(1) Both statement I and statement II are incorrect

(2) Statement I is correct and statement II is incorrect.

(3) Statement I is incorrect and statement II is correct.

(4) Both statement I and statement II are correct.

4) Which of the following organisms do not have membrane bound organelles?
(a) Archaebacteria, (b) Eubacteria, (c) Fungi,
(d) Protozoa, (e) Blue-green bacteria

(1) a, c, d, e

(2) a, b, c

(3) a, b, d

(4) a, b, e

5) Oogamous type of sexual reproduction is shown by

(1) Eudorina

(2) Volvox

(3) Ulothrix

(4) Spirogyra

6) Read the following statements and select the correct option from options given below :-
(A) Whittaker classify the unicellular prokaryotes in monera kingdom.
(B) Dianoflagellates show rapid multiplication by binary fission.
(C) Cell wall of diatoms is made up of silica mainly in which cellulose particles are embeded.
(D) Slime mould form plasmodium and fruiting body during unfavourable condition.
(E) Diatoms are the chief 'Producers' in the oceans.

Options :

(1) A, B and E is true

(2) A, B and D is true

(3) A, B, C and D is true

(4) A, B, C and E is true

7) Match the column I with column II.

	Column I		**Column II**
(A)	Pyrenoids	(i)	*Marchantia*
(B)	Horsetails	(ii)	*Conifers*
(C)	Gemma cups	(iii)	*Chlamydomonas*
(D)	Needle like leaves	(iv)	*Equisetum*

(1) (A) - (iv), (B) - (iii), (C) - (ii), (D) - (i)

(2) (A) - (iii), (B) - (i), (C) - (ii), (D) - (iv)

(3) (A) - (iii), (B) - (iv), (C) - (i), (D) - (ii)

(4) (A) - (ii), (B) - (iii), (C) - (i), (D) - (iv)

8) The lipid like steroidal hormones are synthesized by SER in

(1) Bacteria

(2) Mycoplasma

(3) BGA

(4) Animal cell

9) Match the cell organelles given in column-I with their function in column-II and select the correct option given below:

	Column-I (Cell organelle)		**Column-II (Function of cell organelle)**
A.	Golgi body	P.	Protein synthesis
B.	Ribosome	Q.	Formation of glycoprotein
C.	Lysosome	R.	Hydrolysis of macro molecules
D.	Mitochondria	S.	Oxidation of food

(1) A:Q, B:P, C:S, D:R

(2) A:Q, B:P, C:R, D:S

(3) A:P, B:Q, C:R, D:S

(4) A:P, B:Q, C:S, D:R

10) **Assertion :** The endomembrane system includes endoplasmic reticulum (ER), Golgi complex, lysosomes and vacuoles.

Reason : Mitochondria, chloroplast and peroxisomes are not the part of endomembrane system because their functions are not coordinated with the same.

(1) Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Assertion & Reason are False but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.



11) A stage of cell division is shown in the figure. Select the correct match :-

(1)	Metaphase	Nuclear membrane disintegration
(2)	Pachytene	Crossing over in homologous chromosomes.
(3)	Anaphase	Splitting of centromere & movement of daughter chromosomes to poles
(4)	Diakinesis	Terminalisation of chiasmata

(1) 1

(2) 2

(3) 3

(4) 4

12) Which one of the following stage is correctly described with its event.

(1)	Anaphase I	The bivalent chromosomes align on the equatorial plate
(2)	Diakinesis	Beginning of dissolution of synaptonemal complex.
(3)	Metaphase	Spindle fibres attaches to kinetochore of chromosomes
(4)	Zygotene	Appearance of recombination nodule

(1) 1

(2) 2

(3) 3

(4) 4

13) During photosynthesis, fixation of single CO_2 molecule in maize (A) and tomato (B) plants requires :-

(1) (A) 5 ATP & 2 $NADPH_2$
(B) 3 ATP & 2 $NADPH_2$

(2) (A) 3 ATP & 2 $NADPH_2$
(B) 3 ATP & 2 $NADPH_2$

(3) (A) 2 ATP & 2 $NADPH_2$
(B) 3 ATP & 2 $NADPH_2$

(4) (A) 30 ATP & 12 $NADPH_2$
(B) 18 ATP & 12 $NADPH_2$

14) First reaction in photosynthesis is:

(1) Photolysis of water

(2) Excitation of chlorophyll molecule

(3) Formation of ATP

(4) Fixation of CO_2

15) How many of the following respiratory reactions occur in cytoplasm of eukaryotic cell ?
(a) PGAL → 1, 3 di PGA
(b) Succinyl CoA → Succinic acid
(c) Pyruvic acid → Acetyl CoA
(d) PEP → Pyruvate
(e) F 1, 6 di phosphate → DHAP + 3PGAL

(1) (a) and (d) only

(2) (a), (d) and (e) only

(3) (a), (b), (c) and (e) only

(4) (a) and (e) only

16) In which one of the following processes, CO_2 is not released ?

(1) Aerobic respiration in plants

(2) Aerobic respiration in animals

(3) Alcoholic fermentation

(4) Lactate fermentation

17) How many statement about auxin are correct
(A) Auxin promote apical dominance
(B) Auxin like NAA (Napthalene Acetic Acid) and 2–4–D, (2–4 dichlorophenoxyacetic) have been isolated from plant.
(C) Auxins promote flowering in pineapples
(D) Auxin induce parthenocarpy

(1) One

(2) Two

(3) Three

(4) Four

18) Match the name of scientists with their achievements

	Scientist		Contribution
a.	E-Kurosawa	i.	Isolated auxin from coleoptile tip of oat seedling
b.	Miller et al	ii.	Confirmed the release of volatile substance from ripened oranges that hastens ripening of bananas
c.	F.W. Went	iii.	The foolish seedling disease of rice
d.	H.H. Cousins	iv.	Identified and crystallised kinetin

(1) a–iii, b–ii, c–iv, d–i

(2) a–iii, b–iv, c–i, d–ii

(3) a–iii, b–i, c–iv, d–ii

(4) a–iii, b–i, c–ii, d–iv

19)

Read these statements carefully :-
(i) Cleistogamy
(ii) Anther and stigma of an flower mature in bud condition
(iii) When both of the sex organs mature at same time in a flower.
(iv) Flowers having male and female reproductive structures.

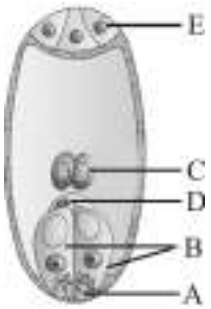
The above contrivances favour:-

(1) Allogamy

(2) Autogamy

(3) Anemophilly

(4) Entomophily

20) 

In the above diagram which of the following is correct function of the corresponding labelling:

(1) D = Formation of primary endosperm

(2) B = Entry of pollen tube in embryosac

(3) E = Secrete chemical substance that attract male gametes

(4) C = Formation of zygote after fusing with one of the male gamete

21)

Arrange the following events of sexual reproduction in correct sequence.
(a) Embryogeny
(b) Triple fusion
(c) Microsporogenesis
(d) Pollen pistil interaction

(1) d, b, a, c

(2) d, c, a, b

(3) d, c, b, a

(4) c, d, b, a

22) **Assertion :** Predators keep prey population under control.
Reason : In the absence of predators prey could achieve high population density causing ecosystem instability.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

23) Which of the following statements are correct ?
(A) Pyramid of biomass in ocean is generally upright.
(B) GFC is major conduit of energy flow in terrestrial ecosystem.
(C) Standing crop is measured as the mass of living organisms or the number in a unit area.
(D) Trophic level represents a species not its function.
(E) Molluscs are herbivores in aquatic ecosystem.
Choose correct option :

(1) A, B, C, D

(2) B, C, D

(3) C, D, E

(4) C and E

24) Which statements is/are correct :
(A) In most ecosystem all the pyramids of number, of energy and biomass are upright.
(B) Energy at a lower trophic level is always more than at a higher level.
(C) The pyramid of biomass in sea is generally inverted because the biomass of fishes far exceed that of phytoplankton.
(D) Saprophytes are not given any place in ecological pyramids eventhough they play vital role in the ecosystem.

(1) Only A & C

(2) Only C & D

(3) Only A, B & C

(4) A, B, C & D

25) Read the given statements and select the correct option regarding this.
(i) Ecosystem services provided by nature to human beings such as oxygen for respiration, aesthetic value, etc.
(ii) Direct economic benefits derived from nature by human beings such as food, medicines, etc.
(iii) Every species has an intrinsic value, even if it is not of any economic use to us.
With respect to above given codes (i), (ii) and (iii) choose the correct option.

	Narrowly utilitarian	Broadly utilitarian	Ethical
(1)	(iii)	(ii)	(i)
(2)	(ii)	(i)	(iii)
(3)	(i)	(ii)	(iii)
(4)	(ii)	(iii)	(i)

(1) 1
(2) 2
(3) 3
(4) 4

26) Match the column-I with II :-

Column-I		**Column-II**	
(A)	Dodo	(i)	Africa
(B)	Quagga	(ii)	Bali, Javan Caspian
(C)	Thylacine	(iii)	Mauritius
(D)	Steller's sea cow	(iv)	Australia
(E)	Tiger sub-species	(v)	Russia

	A	B	C	D	E
(1)	iv	ii	i	v	iii
(2)	iii	iv	ii	i	v
(3)	iii	i	iv	v	ii
(4)	iii	v	iv	i	ii

(1) 1
(2) 2
(3) 3
(4) 4

27) **Assertion :-** Tropical environments are less seasonal, more constant and predictable and hence more species diverse.

Reason :- Constant environments promote niche specialisation and lead to greater species diversity.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

28) Which biomolecules is classified in monoglyceride, diglyceride and Triglyceride categories ?

(1) Carbohydrate
(2) Lipid
(3) Protein
(4) Nucleic acid

29) Phosphofructokinase (PFK) is inhibited by :-

(1) ADP
(2) ATP
(3) Fructose-6-P
(4) Fructose -1,6-bi-P

30) The protein portion of the enzyme is called the (A). Three kind of co-factor were identified (B), (C) and (D).

(1) (A)-holo enzyme, (B) Prosthetic group,(C) Co-enzymes, (D) Metal-ions
(2) (A)-Apoenzyme, (B) Prosthetic group, (C) Co-enzymes, (D) Metal-ions
(3) (A)-Co-enzyme, (B) Metal ions, (C) Apoenzyme, (D) Prosthetic group
(4) (A)-Holo enzyme, (B) Apoenzyme, (C) Prosthetic group, (D) Metal-ions.

31) Which of the following statements is false about leaf ?

(1) A leaf is said to be simple, when its lamina is entire or when incised, the incisions do not touch the midrib
(2) A leaf is said to be compound when the incisions of lamina reach upto the midrib breaking into a number of leaflets
(3) Leaf is the most important vegetative organ for photosynthesis
(4) Leaf is not a transpiratory organ

32)

A scar on the seed coat through which the developing seeds are attached to the fruit is called.

(1) Testra
(2) Tegmen
(3) Hilum
(4) Micropyle

33) Select correct combination of column I, II & III.

	Column I	Column II	Column III
(A)	Marginal	 (I)	(1) Sunflower, Marigold
(B)	Axile	 (II)	(2) Dianthus, Primrose
(C)	Parietal	(III) 	(3) Mustard, Argemone
(D)	Free Central	(IV) 	(4) China rose, Tomato, Lemon
(E)	Basal	 (V)	(5) Pea

(1) A-V-(5); B-II-(4); C-I-(3); D-III-(2); E-IV-(1)

(2) A-I-(5); B-II-(4); C-III-(3); D-IV-(2); E-IV-(1)

(3) A-V-(1); B-II-(4); C-I-(2); D-III-(3); E-IV-(5)

(4) A-V-(1); B-III-(2); C-II-(4); D-I-(5); E-IV-(3)

34) In which of the following characters, a monocot root differs from a dicot root?

(1) Radial vascular bundles

(2) Conjuctive tissue in between xylem and phloem

(3) Large pith

(4) Single layered endodermis

35) In grasshopper :-

(a) Total number of chromosomes are same in both ♂ and ♀

(b) Two types of gametes are formed by male

(c) Female heterogamety is present

(1) a, b & c are correct
(2) a & c are correct
(3) a & c are incorrect
(4) a & b are incorrect

36) **Statement-I :** Law of segregation is based on the fact that the alleles do not show any blending and both the characters are recovered as such in F_1 generation.
Statement-II : Law of dominance is used to explain the expression of only one of parental character in a monohybrid cross in the F_1 and expression of both in the F_2.

(1) Both Statement-I and Statement-II are correct.
(2) Statement-I is correct and Statement-II is incorrect.
(3) Statement-I is incorrect and Statement-II is correct.
(4) Both Statement-I and Statement-II are incorrect.

37) What is true for Mendel ?

(1) Qualitative study of qualitative characters
(2) Quantitative study of quantitative characters
(3) Qualitative study of quantitative characters
(4) Quantitative study of Qualitative characters

38) A pure tall pea plant is crossed with pure dwarf pea plant and their progeny is self pollinated. The ratio of true tall and dwarf plants will be -

(1) 3 : 1
(2) 1 : 3
(3) 1 : 1
(4) 1 : 0

39) Select false statement for independent assortment :-

(1) Segregation of one pair of unit factor is independent of the other pair of unit factor.
(2) During meiosis-I the two chromosome pairs can align at the metaphase plate independently of each other.
(3) Due to independent assortment a dihybrid produce 4 types of gametes in the ratio of 9 : 3 : 3 : 1.
(4) Gene of different characters are located on different pair of chromosome.

40)

Match the column :-

	A		**B**
(A)	rRNA	(i)	Peptide bond formation

(B)	tRNA	(ii)	Codon identification
(C)	SnRNA	(iii)	Splicing
(D)	mRNA	(iv)	Genetic information

Option :

(1) A-i, B-ii, C-iii, D-iv

(2) A-ii, B-i, C-iii, D-iv

(3) A-iii, B-i, C-ii, D-iv

(4) A-iv, B-iii, C-ii, D-i

41) **Assertion (A) :** Sometime the regulatory sequences are loosely defined as regulatory genes, even though these sequences do not code for any RNA or protein.
Reason (R) : Inheritance of a character is also affected by promoter and regulatory sequences of a structural gene.

(1) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are correct and **(R)** is not the correct explanation of **(A)**.

(3) **(A)** is correct but **(R)** is incorrect.

(4) Both **(A)** and **(R)** are correct.

42) **Statement-1 :** DNA fingerprinting is a very time consuming and tedious way to compare DNA sequences of two individuals.
Statemenet-2 : DNA fingerprinting is the basis of paternity testing, in case of disputes.

(1) Statement-1 and 2 both correct

(2) Statement-1 and 2 both incorrect

(3) Statement-1 is correct 2 incorrect

(4) Statement-1 is incorrect 2 correct

43) In Eukaryotes, which of the following mechanism is not correct in terms of regulation of gene expression ?

(1) Transcription

(2) Splicing

(3) Transport of mRNA from cytoplasm to nucleus

(4) Translation

44) In 1953, Watson and Crick proposed DNA double helix model and won the noble prize. Their model of DNA was based on :-
(A) X ray diffraction studies of DNA done by Wilkins and Franklin.
(B) Chargaff's base equivalence rule.
(C) Griffith's transformation experiment.
(D) Meselson and Stahl's experiment.

(1) A, B and D

(2) A and B

(3) A, B, C and D

(4) B and D

45) How many statements are **correct** :-
(a) VNTR belong to class of a minisatellite.
(b) Satellite DNA do not code for proteins.
(c) DNA from every cell or tissue from an individual show different degree of polymorphism.
(d) 99.9 percent DNA of all humans are same

(1) All statements

(2) Only one statement

(3) 2 statements

(4) 3 statements

46) A small group of worm-like marine animals with a cylindrical body divided into a proboscis, collar, and trunk belongs to :

(1) Platyhelminthes

(2) Annelida

(3) Hemichordata

(4) Aschelminthes

47) Marine animals characterized by the presence of comb plates belong to the phylum :

(1) Echinodermata

(2) Ctenophora

(3) Annelida

(4) Mollusca

48) Match the phylum with its distinctive character and select the correct option.

	Phylum		**Character**
(i)	Porifera	(a)	Jointed appendages
(ii)	Coelenterata	(b)	Water vascular system
(iii)	Arthropoda	(c)	Choanocytes
(iv)	Echinodermata	(d)	Cnidoblasts

(1) i-d, ii-c, iii-b, iv-a

(2) i-c, ii-d, iii-a, iv-b

(3) i-c, ii-a, iii-d, iv-b

(4) i-b, ii-d, iii-c, iv-a

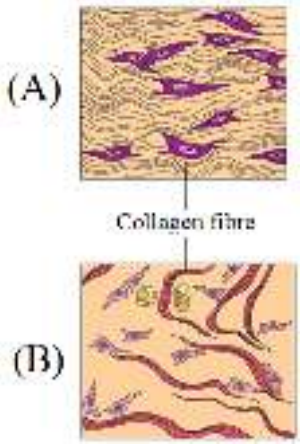
49) **Assertion (A) :-** Bony fishes can do rest when they are tired while swimming.
Reason (R) :- They have air bladder.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.



50) Select the correct statement about the given diagram

(1) In **B**-collagen fibres are present in rows between many parallel bundles of fibres

(2) In **A**-fibroblast and many fibres are oriented differently

(3) Example of **A** is Tendon

(4) **B** is an example of specialised C.T.

51) In cockroach head is connected with thorax by a short extension of _____ known as neck.

(1) Prothorax

(2) Mesothorax

(3) Metathorax

(4) Abdomen

52)

Read the following statements about the frog and choose the correct option :
(A) The undigested solid waste moves into the rectum and passes out through anus.
(B) RBC's are enucleated and contain haemoglobin.
(C) The ventricle opens into a sac like conus arteriosus.
(D) It is an ammonotelic animal.
(E) On land the buccal cavity, skin and lungs act the respiratory organs.

(1) All are incorrect

(2) Only A, B, C and D are incorrect

(3) Only D is incorrect

(4) Only A, B and D are incorrect

53) Match the following columns :

Column-I		**Column-II**	
(i)	Diastolic blood pressure	(a)	120 mm Hg

(ii)	Angina	(b)	Atherosclerosis
(iii)	C.A.D.	(c)	80 mm Hg
(iv)	Systolic blood pressure	(d)	Acute chest pain

(1) i-a, ii-d, iii-b, iv-c

(2) i-a, ii-c, iii-d, iv-b

(3) i-c, ii-d, iii-b, iv-a

(4) i-b, ii-c, iii-d, iv-a

54) Match the columns :

	Column-A		**Column-B**
i.	I^{st} heart sound	a.	Atrial depolariation
ii.	Stroke vol.	b.	Brady cardia
iii.	P wave	c.	Blood ejected by each ventricle per beat
iv.	Vagus	d.	Closure of AV valves

(1) i-a, ii-b, iii-c, iv-d

(2) i-b, ii-c, iii-a, iv-d

(3) i-d, ii-c, iii-a, iv-b

(4) i-d, ii-a, iii-b, iv-c

55) **Statement-I** : Ventricular contraction starts shortly after Q.
Statement-II : Any deviation in the shape of ECG indicates a possible abnormalities.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is incorrect but **Statement II** is correct.

(3) Both **Statement I** and **Statement II** are correct.

(4) **Statement I** is correct but **Statement II** is incorrect.

56) Read the following statements:-
(a) Fibrinogens are needed for clotting or coagulation of blood.
(b) Factors for clotting of blood are also present in the plasma in an active form.
(c) Haemoglobin play a significant role in transport of respiratory gases.
(d) Basophills (2-3 percent) resist infection and are also associated with allergic reactions.
(e) Blood normally contains 1,50000-3,50000 platlets mm^{-3}.
Choose the correct options:-

(1) a, b and d

(2) b, c and e

(3) a, c and e

(4) c, d and e

57) Which of the following factors are responsible for the left shift in the oxygen dissociation curve :-

(1) High pO_2, low pCO_2, low pH, high $[H^+]$

(2) High pCO_2, low pO_2, low temperature, high pH

(3) High pO_2, low temperature, high pH, low concentration of intermediate respiratory produces

(4) High pCO_2, low pO_2, high temperature, low pH

58) These are two statements given.
Statement-I :- The minimum volume of air accomodated in the lungs at the end of a forced expiration is called residual volume.
Statement-II :- The maximum volume of air a person can breath out after forceful inspiration is called vital capacity of lungs.
Choose the correct option :-

(1) Statement-I and statement-II both are false.

(2) Statement-I is true but statement-II is false.

(3) Statement-I and statement-II both are true.

(4) Statement-I is false but statement-II is true.

59) See carefully some factors given below :
A. High Temperature B. High pH
C. High pCO_2 D. High pO_2
E. Low pH F. Low Temperature
G. Low pCO_2 H. Low pO_2
Which factors given above favours dissociation of oxyhaemoglobin at tissue level ?

(1) A, C, E, H

(2) A, C, E, F

(3) A, C, G, H

(4) A, C, E, G

60) Removal of proximal convoluted tubule from the nephron will result in :-

(1) More concentrated urine

(2) No change in quality and quantity of urine

(3) No urine formation

(4) More diluted urine

61) **Assertion (A) :** Shark are said to be ammoniotelic animals and kidney play significant role in its removal.
Reason (B) : Shark have ability to retain considerable amount of ammonia in vertebrates.

(1) Both (A) & (R) are true and (R) is not correct explanation of (A)

(2) Both (A) & (R) are true and (R) is correct explanation of (A)

(3) Both (A) and (R) are false

(4) (A) is true but (R) is false

62) Given below are two statements : One is labelled as Assertion A and the other is labeled as Reason R :

Assertion (A) : Each kidney has nearly one million complex tubular structures called nephron.
Reason (R) : Each nephron has two parts glomerulus and the Malpighian tubules

(1) Both A and R true and R is not the correct explanation of A

(2) A is true but R is false

(3) A is false but R is true

(4) Both A and R true but R is the correct explanation of A

63) Read the following terminology of column I & II and identify the correct match :

	Column I		**Column II**
(1)	Neurohypophysis	(a)	TSH
(2)	Adenohypophysis	(b)	Oxytocin
(3)	CNS	(c)	Vasopressin
(4)	Neuron	(d)	Brain and spinal cord
		(e)	Cell body, dendrites and axon
		(f)	ACTH

(1) 1-a,f; 2-b,f; 3-e,f; 4-d

(2) 1-b,c; 2-a,f; 3-d; 4-e

(3) 1-b,c; 2-a,f; 3-e; 4-d

(4) 1-e; 2-b,c; 3-a,f; 4-d

64) Read the following terminology of column-I and column-II and identify their correct match:-

Column-I		**Column-II**	
(A)	Parathyroid harmone	(i)	Gap between pre and post syoptic membrane
(B)	Aldosterone	(ii)	Inner most meninges in contact with brain
(C)	Piamater	(iii)	Main mineralocorticoid in out body
(D)	Synaptic left	(iv)	Hypercalcemia hormone

(1) A-(i), B-(iv), C-(iv), D-(iii)

(2) A-(ii), B-(iii), C-(iv), D-(i)

(3) A-(iv), B-(iii), C-(ii), D-(i)

(4) A-(i), B-(ii), C-(iii), D-(iv)

65) Q. Given below are two statements one is labelled as Assertion(A) and the other is labelled as Reason(R).
Assertion (A) : When we do physical exercises, the energy demand is increased for maintaining an

increased muscular activity. The supply of oxygen is also increased. The increased supply of oxygen necessitates an increase in the rate of respiration
Reason (R) : The functions of organs in our body must be coordinated to maintain homeostasis.

(1) Both (A) and (R)are correct and (R)is not the correct explanation of (A)

(2) (A) is not correct but (R)is correct

(3) (A) is correct but (R)is not correct

(4) Both (A) and (R)are correct and (R)is the correct explanation of (A)

66) Which hormones do stimulate the production of bicarbonate and pancreatic juice respectively ?

(1) Cholecystokinin and secretin

(2) Insulin and glucagon

(3) Angiotensin and epinephrine

(4) Secretin and Cholecystokinin

67) Read the following statements and choose the correct set of statements :
(a) Insulin enhances cellular glucose uptake and utilisation.
(b) Prolonged hyperglycaemia leads to a complex disorder called diabetes insipidus.
(c) The endocrine pancreas consists of "Islets of Langerhans'.
(d) Glucagon acts mainly on the liver cells (hepatocytes) an stimulates glycogenolysis.
Choose the correct answer from the options given below :

(1) (a) and (c) only

(2) (a), (b), (c) and (d) only

(3) (a), (c) and (d) only

(4) (a) and (d) only

68)

Match the Column I with Column II & Column III and choose the correct option

Column I		**Column II**		**Column III**	
(A)	Prolactin	(i)	Stimulate estrogen secretion	(a)	Lactation hormone
(B)	TSH	(ii)	Growth of mammary glands	(b)	Increased in simple goitre
(C)	ACTH	(iii)	Stimulate cortisol secretion	(c)	Used in infertility treatment
(D)	FSH	(iv)	Stimulate thyroxin secretion	(d)	Regulated by CRH

(1) A (ii) (c), B (iii) (d), C (i) (b), D (iv) (a)

(2) A (i) (d), B (ii) (a), C (iv) (b), D (iii) (c)

(3) A (ii) (a), B (iv) (b), C (iii) (d), D (i) (c)

(4) A (ii) (a), B (iv) (d), C (i) (b), D (iii) (c)

69) Given below are two statements one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Androgens regulate the development, maturation and functions of the male accessory sex organs like epididymis, vas deferens, seminal vesicles, prostate gland, urethra etc.
Reason (R) : Androgens act on the central neural system and influence the male sexual behaviour (libido).
In the light of above statements, choose **the most appropriate answer** from the options given below :

(1) Both **(A)** and **(R)** are correct and **(R)** is not the correct explanation of **(A)**.

(2) **(A)** is incorrect but **(R)** is correct.

(3) **(A)** is correct but **(R)** is incorrect.

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**.

70) Cardiac muscle are the muscle of the heart. They are _____ , _____ and _____.

(1) Striated, branched, involuntary

(2) Excitable, contractable, smooth

(3) Smooth, contractable, elastic

(4) Striated, branched, voluntary

71) Name the molecule responsible for the detachment of myosin from actin during mechanism of muscle contraction.

(1) ADP

(2) ATP

(3) CP (Creatine phosphate)

(4) GTP

72) Glenoid & Acetabulum are related with -

(1) Pivot joint

(2) Hinge joint

(3) Ball & socket joint

(4) Glidding joint

73) onsider the following four statements (a-d) and select the option which includes all the incorrect ones only.
(a) Each testicular lobule contains 750 highly coiled seminiferous tubules, in which sperms are produced.
(b) Sertoli cells undergo meiotic divisions finally leading to sperm formation.
(c) The regions outside the seminiferous tubules called interstitial spaces, contain small blood vessels, Leydig cells, and some immunologically competent cells.
(d) The scrotum helps in maintaining the low temperature of the testes, necessary for spermatogenesis.

(1) Statements (b), (c) and (d).

(2) Statements (a), (b).

(3) Statements (a), (b) and (c).

(4) Statements (a), (c) and (d).

74) Which of the following statements are correct?
(a) Fertilisation can occur only if the ovum and sperm are transported simultaneously to the ampulla.
(b) By the end of the second month of pregnancy the first movements of foetus can be observed.
(c) On the 14th day of menstrual cycle, ovulation takes place.
(d) Lack of menstruation is always an indication of pregnancy.
Options:

(1) a and c

(2) b and c

(3) c and d

(4) a and b

75) **Assertion:** Mammary glands of the female undergo differentiation during pregnancy.
Reason: Hormonal changes during pregnancy prepares the mammary glands to produce milk for child care.

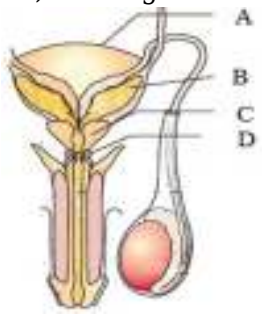
(1) Both assertion and reason are true and reason is correct explanation of assertion.

(2) Both assertion and reason are true but reason is not correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false.

76) In the given diagram identify A, B, C and D respectively and choose the correct option:



	A	B	C	D
(1)	Epididymis	Prostate gland	Vas deferens	Bulbourethral gland
(2)	Urinary bladder	Prostate gland	Testis	Bulbourethral gland
(3)	Urinary bladder	Prostate gland	Seminal vesicle	Bulbourethral gland
(4)	Urinary bladder	Seminal vesicle	Prostate gland	Bulbourethral gland

(1) 1

(2) 2

(3) 3

(4) 4

77) Match the following :

A	Non- medicated IUDs	a	Multiload 375	i	Uterus unsuitable for implantation
B	Copper Releasing IUDs	b	Saheli	ii	Phagocytosis of sperm
C	Hormone Releasing IUDs	c	Lippes loop	iii	Suppress sperm motility
D	Weekly oral pill	d	LNG-20	iv	Anti-estrogenic activity

(1) A-b-i, B-c-ii, C-d-iii, D-a-iv

(2) A-c-iv, B-a-iii, C-b-i, D-d-ii

(3) A-c-ii, B-a-iii, C-d-i, D-b-iv

(4) A-a-iii, B-c-ii, C-d-i, D-b-iv

78) Which of the following explanation is/are related with bone marrow and thymus ?
(a) Act as a filter of the body by trapping blood borne micro-organisms.
(b) Provide micro-environment for the development and maturation of T-lymphocytes.
(c) Immature lymphocytes differentiate into antigen sensitive lymphocytes.
(d) Provide the sites of interaction of lymphocyte with the antigen.

(1) a, b, c & d

(2) a, b & c only

(3) b & c only

(4) c only

79) Colostrum secreted by mother provides -

(1) Natural active immunity

(2) Artificial active immunity

(3) Natural passive immunity

(4) Artificial passive immunity

80) **Statement-I :** The thymus is a lobed organ located near heart and above the breastbone.
Statement-II : The primary lymphoid organ where mature lymphocytes differentiate into antigen-sensitive lymphocytes.

(1) Both statement-I and statement-II are true.

(2) Both statement-I and statement-II are false.

(3) Statement-I is true but statement-II is false.

(4) Statement-I is false & statement-II is true.

81) Read the following disease carefully -
Typhoid, common cold, dengue, malaria, pneumonia, Hepatitis-B, Plague, AIDS, Tuberculosis
How many of the above infections are caused by virus ?

(1) Two

(2) Three

(3) Four

(4) Five

82) In millers experiment in 1953, which molecules were formed ?

(1) DNA and RNA

(2) Amino acids

(3) Enzymes

(4) Vitamins

83) Which of the following is a non-medicated IUD?

(1) Lippes loop

(2) Multiload 375

(3) Progestasert

(4) LNG-20

84) Read the following Assertion (A) and Reason (R) and choose the correct option
Assertion (A): When reptiles came down mammals took over this earth.
Reason (R): Mammals were viviparous and protected their unborn young inside the mother's body.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is incorrect

(4) Both (A) and (R) are incorrect

85) Fill in the blanks :
The essence of Darwinian theory about evolution is ____a____. The rate of appearance of new form is linked to the ___b___. Fitness is based on characteristics which are ___c___. There must be a ___d___ for selected and to evolve.

(1) a - Natural selection, b - Life cycle
c - Acquired, d - Genetic basis

(2) a - Natural selection, b - Life cycle
c - Genetic basis, d - Acquired

(3) a - Natural selection, b - Life span
c - Anthropogenic, d - Genetic basis

(4) a - Natural selection, b - Life span
c - Inherited, d - Genetic basis

86) **Assertion :-** ELISA is based on the principle of antigen-antibody interaction.
Reason :- Infection by pathogen can be detected by the presence of antigens or by the detecting the antibody synthesised against the pathogen.

(1) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(2) Assertion is True but the Reason is False.

(3) Assertion is False but the Reason is True.

(4) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

87) **Statement-I :** Each restriction endonuclease functions by inspecting the length of a DNA sequence.
Statement-II : Each restriction endonuclease recognises a specific palindromic nucleotide sequences in the DNA.

(1) Both Statement-I and Statement-II are true

(2) Both Statement-I and Statement-II are false

(3) Statement-I is true but statement-II is false.

(4) Statement-I is false but statement-II is true.

88) **Statement-I :-** The first transgenic cow, Rosie, produced human protein-enriched milk (2.4 grams per litre).
Statement-II :- The milk contained the human protein (α-1antitrypsin) used to treat emphysema.

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

89) Some strains of Bacillus thuringiensis produce proteins that kill certain insects such as :-

(1) Lepidopterans

(2) Coleopterans

(3) Dipterans

(4) All of the above

90) Match List-I with List-II

List-I		**List-II**	
A.	Monascus purpureus	I.	Citric acid
B.	Aspergillus niger	II.	Cyclosporin A
C.	Acetobacter aceti	III.	Statins
D.	Trichoderma polysporum	IV.	Acetic acid

Choose the correct answer from the options given below :

(1) A-III, B-I, C-II, D-IV

(2) A-II, B-IV, C-III, D-I

(3) A-III, B-I, C-IV, D-II

(4) A-IV, B-I, C-III, D-II

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	4	2	3	4	4	1	4	3	2	4	2	1	3	3	2	3	2	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	4	1	2	3	2	1	2	3	2	2	3	4	1	2	3	3	3	3	3
Q.	41	42	43	44	45															
A.	1	4	3	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	1	3	3	3	1	3	1	2	2	1	3	3	2	3	3	4	1	3	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	4	4	4	1	3	3	4	4	1	3	4	1	3	4	3	3	3	4	3
Q.	86	87	88	89	90															
A.	1	2	1	2	2															

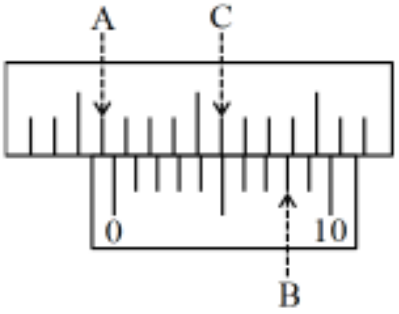
BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	1	4	4	2	1	3	4	2	1	3	3	1	2	2	4	3	2	2	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	1	4	4	2	3	1	2	2	2	4	3	1	3	3	3	4	3	3	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	4	3	2	4	3	2	2	4	3	1	4	3	3	3	3	3	3	1	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	2	3	4	4	3	3	1	1	2	3	2	1	1	4	3	3	3	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	2	1	1	4	4	1	3	4	3										

SOLUTIONS

PHYSICS

1) 9MSD = 10VSD ⇒ 1VSD = 0.9MSD = 0.9 mm
Here the coinciding division of Vernier is 5th, hence



Distance AB = AC + CB
= 5MSD + 3VSD
= 5mm + 3 × 0.9 mm = 7.7 mm

2) $Y = (5 \times 10^4) \times \frac{M}{\ell}\frac{dyne}{cm^2}$
$\Rightarrow \frac{\Delta Y}{Y} \times 100\% = \left(\frac{\Delta M}{M} + \frac{\Delta \ell}{\ell}\right) \times 100\%$
$= \left(\frac{20}{500} + \frac{0.08}{4}\right) \times 100\%$
= 4 + 2 = 6%

3) $V_T \propto R^2 \Rightarrow$ A is incorrect
For a liquid T↑ η↓ v_T↑
V_T does not depend upon initial velocity ⇒ C is incorrect
$p = mV_T \Rightarrow p \propto (R^3)(R^2) \Rightarrow p \propto R^5 \Rightarrow$ D is correct

4) Let mass of A = 3m, mass of B = 5m
Specific heat of A = 4s, specific heat of B = 3s
Heat lost = heat gained
(3m) (4s) (84–T) = (5m)(3s) (T–24)
⇒ 1008–12T = 15T – 360
⇒ 1368 = 27T ⇒ T = 50°C

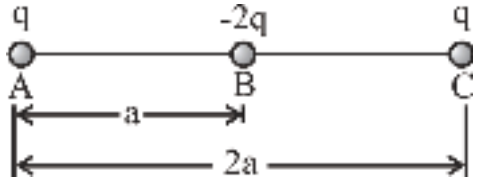
5) From graph : $R_{eq} = \frac{V}{I} = \frac{8}{4mA} = 2000\Omega$ ...(1) From circuit : $R_{eq} = \frac{R(12000)}{12000 + R}$ ...(2)
From (1) & (2) : $\frac{R(12000)}{12000 + R} = 2000$
⇒ 6R = 12000+R
$\Rightarrow R = \frac{12000}{5} = 2400\Omega$

6) At minimum deviation ∠i = ∠e; A ⇒ correct
There are two values of incidence for which deviation is same except ∠i = ∠e; B ⇒ correct
Larger angle prism may provide more deviation due to larger path inside prism; C ⇒ incorrect
At minimum deviation 2r = A ⇒ r = A/2;
D ⇒ incorrect

7) Let $R_1 = 50\Omega$, $V_{max} = 10V$, $V_Z = 6V$
In zener diode $\frac{V_{max} - V_Z}{R_1} \geqslant I_{knee} + I_R$
$\frac{(10-6)}{50} \geq 5mA + I_R \Rightarrow I_R \leq 80mA - 5mA$
$I_{R(max)} = 75mA$ (maximum current through R)
So, the minimum value of resistance R is
$$R_{min} = \frac{V}{I_{R(max)}} = \frac{6}{75mA} = \frac{6000}{75} = 80\Omega$$

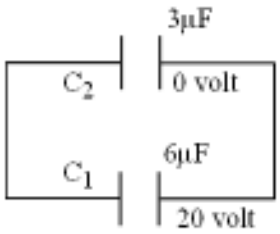
8) $\vec{\tau} = \vec{p} \times \vec{E}$
$\vec{\tau} = \left(2\hat{i} + 3j\right) \times \left(4\hat{i}\right)$
$\vec{\tau} = 2\left(\hat{i} \times \hat{i}\right) + 3 \times 4\left(\hat{j} \times \hat{i}\right)$
$\vec{\tau} = 0 + 12\left(-\hat{k}\right)$
$\vec{\tau} = -12\hat{k}\,N-m$

9) Acceleration of +3Q = $\frac{3QE}{m}$ (↓)
Acceleration of –2Q = $\frac{2QE}{m}$ (↑)

10) 

$U_{AB} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{(q)(-2q)}{a}$
$U_{BC} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{(-2q)(q)}{a}$
$U_{CA} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{(q)(q)}{2a}$
$\therefore U_{system} = \frac{1}{4\pi\varepsilon_0} \cdot \left[\frac{-2q^2}{a} - \frac{2q^2}{a} + \frac{q^2}{2a}\right] U_{system} = -\frac{7q^2}{8\pi\varepsilon_0 a}$

11)



$$V_C = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$

$$V_C = \frac{6 \times 20 + 3 \times 0}{3 + 6} = \frac{40}{3}\text{volt}$$

$$q_2^1 = C_2V_C = 3 \times \frac{40}{3} = 40\mu C$$

$q_2' = C_2 V_2$ ....

12)

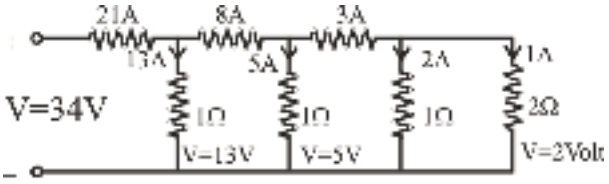
By maximum power transfer theorem for maximum power in y

R + 2 = y

R + 2 = 5

R = 3Ω

13)



14)

A. Given: Current (I) = 10 A. Area (A) = $1cm^2 = 1 \times 10^{-4}m^2$.

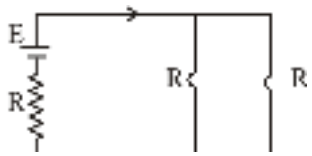
B. Density (n) = $9 \times 10^{28}\, m^{-3}$. Charge of electron (e) = 1.6×10^{-19} C.

C. Formula: $v_d = \frac{I}{nAe}$.

D. $$v_d = \frac{10}{\left(9 \times 10^{28}\right) \times \left(1 \times 10^{-4}\right) \times \left(1.6 \times 10^{-19}\right)}.$$

E. $$v_d = \frac{10}{14.4 \times 10^5} \approx 6.94 \times 10^{-6}\text{m/s}.$$

15)



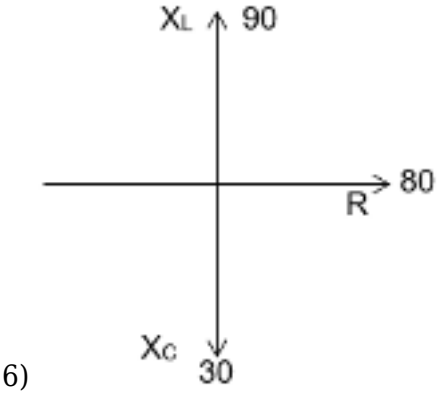
E = B□V

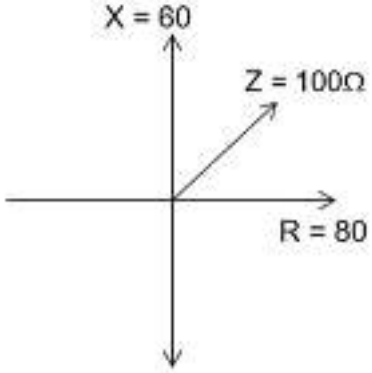
$$I = \frac{\varepsilon}{R_{eq}}$$

$$I = \frac{\varepsilon}{3R/2} = \frac{2\varepsilon}{3R}$$

$I = \frac{2B\ell V}{3R}$, $I_1 = I_2 = \frac{I}{2} = \frac{B\ell V}{3R}$

16)





$Z = \sqrt{X^2 + R^2}$

where $X = |X_L - X_C|$

$\Rightarrow Z = 100\Omega$

and $i_{rms} = \frac{V_{rms}}{Z} = \frac{200\,V}{100\Omega} = 2A$

and power discipated in the coil

$P = i_{rms}^2 R_{coil} = (2)^2 \cdot (36) = 144\ W$

17) $(U_{av})_B = (U_{av})_E = \frac{1}{4}\in_0 E_0^2$

$\frac{1}{4}\in_0 (2)^2 = 8.85 \times 10^{-12} J/m^3$

18)

Since the particle exists the magnetic field region with a velocity $\vec{v} = \left(3\hat{i} + 3\sqrt{3}\hat{j}\right)$ m/s

$\tan\theta = \frac{3\sqrt{3}}{3}$

$\therefore \theta = \frac{\pi}{3}$

The angular displacement of the particle in the magnetic field region is $\theta = \frac{\pi}{3}$

Angular velocity, $\omega = \frac{qB}{m}$

$\therefore t = \frac{\theta}{\omega} = \frac{\pi m}{3qB}$

19)

Magnetic field on axis at distance x :

$$B = \frac{B_0}{(1 + x^2/R^2)^{3/2}}$$

Given : $B = \frac{B_0}{8}$

So, $\frac{B_0}{(1 + x^2/R^2)^{3/2}} = \frac{B_0}{8}$

Cancelling B_0,

$$\left(1 + \frac{x^2}{R^2}\right)^{3/2} = 8$$

Taking power $\frac{2}{3}$ on both sides,

$$1 + \frac{x^2}{R^2} = 4$$

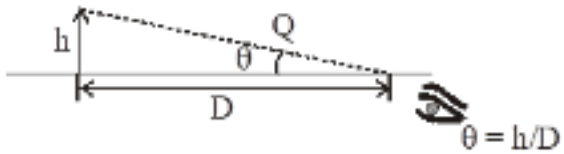
So, $\frac{x^2}{R^2} = 3$; $x = R\sqrt{3}$

20) $V_A = \frac{uf}{u-f} = \frac{(-25) \times 10}{(-25)-10} = \frac{-250}{-35} = \frac{50}{7}$cm

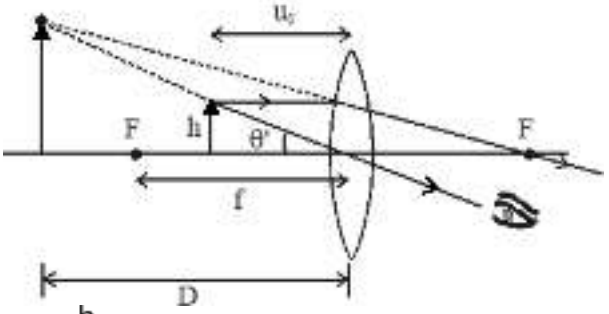
$V_B = \frac{fu}{u-f} = \frac{10 \times (-20)}{-20-10} = \frac{-200}{-30} = \frac{20}{3}$

length of image = $\frac{50}{7} - \frac{20}{3} = \frac{10}{21}$ cm

21)



$\theta = h/D$



$\theta' = \frac{h}{u_0}$; θ' is same for both object and image

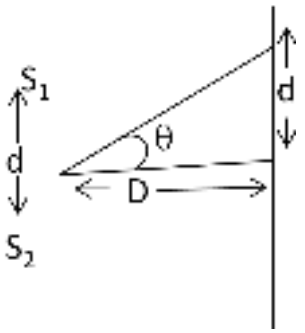
$m = \frac{\theta'}{\theta} = \frac{D}{u_0}$

$u_0 < D$

Hence $m > 1$

22)

For dark fringe $\Delta x = (2n-1)\frac{\lambda}{2}$



$\Delta x = d\sin\theta = (2n-1)\frac{\lambda}{2}$

$d\left(\frac{d/2}{D}\right) = (2n-1)\frac{\lambda}{2}$

$\frac{d^2}{2D} = (2n-1)\frac{\lambda}{2} \Rightarrow \lambda = \frac{d^2}{(2n-1)D}$

n = 1,2,3 .....

$\lambda = \frac{d^2}{D}, \frac{d^2}{3D}, \frac{d^2}{5D}$

23)

$h\nu = \phi + 1.2$ eV ...(1)

$\frac{3h\nu}{2} = \phi + 3.6$ eV ...(2)

divide 2 by 1 and solve,

We get $\phi = 3.6$ eV.

24) $R = n^2a_0$

$2.12 = n_1^2 \times 0.53$

$n_1 = 2$

$4.77 = n_2^2 \times 0.53$

$n_2 = 3$

$E = 13.6\left(\frac{1}{4} - \frac{1}{9}\right) = \frac{13.6 \times 5}{36} = \frac{68}{36} = 1.89$eV

25)

For an electron moving in a circular path in a transverse magnetic field

Centripetal force = Magnetic force

$mv^2/r = evB$

$mv = eBr$

The momentum p of the electron is

$p = mv = eBr$

The de-Broglie wavelength is given by

$\lambda = h/p$

$\lambda = h/(eBr)$

26)

E = N (energy per fission)

$10^3 = N(200 \times 10^6 \times 1.6 \times 10^{-19})$

$$\Rightarrow N = \frac{10^3}{200 \times 10^6 \times 1.6 \times 10^{-19}}$$

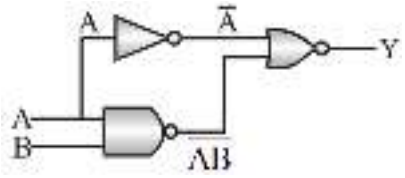
$N = 3.1 \times 10^{13}$.

27)

$$\text{Total No. of Nuclei} = \frac{6.023 \times 10^{23}}{235} \times 2 \times 10^3$$

$$P_{out} = \frac{6.023 \times 10^{23} \times 2 \times 10^3 \times 185 \times 10^6 \times 1.6 \times 10^{-19}}{235 \times 30 \times 24 \times 60 \times 60}$$

$P_{out} = 58.46$ MW

28) 

$$Y = \bar{A} + \overline{A \cdot B} = \bar{A} \cdot \overline{\overline{A \cdot B}} = A.A.B = A.B$$

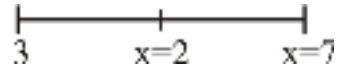
29)

$U = 2 - 20x + 5x^2$

$$F = -\frac{dU}{dx} = -(-20 + 10x)$$

$F = 20 - 10x$

x = 2 M.P.



A = 5 cm $x_{max} = 7$ m

30)

	A	B
Frequency	n_A	n_B
Tension	T	T
Length	2L	3L
Radii	3r	2r

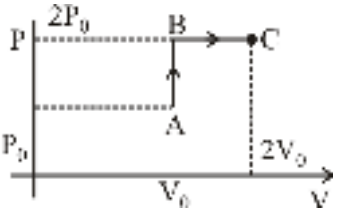
Vibration Fundamental Second Overtone

$$n_A = \frac{1}{2(2L)}\sqrt{\frac{T}{\rho\,\pi\,(3r)^2}}$$

$$n_B = \frac{3}{2(3L)}\sqrt{\frac{T}{\rho\pi(2r)^2}}$$

$$\frac{n_A}{n_B} = \frac{1}{3}$$

31)



Let initial state A: P^0, V^0, $T_A = \frac{P_0V_0}{nR}$

Step 1 : constant volume A → B, P doubles:

$P_B = 2P_0$, $V_B = V_0$

$$T_B = \frac{(2P_0)V_0}{nR} = 2T_A$$

Step 2: Constant pressure B → C, V doubles:

$V_C = 2V_0$, $P_C = 2P_0$.

$$T_C = \frac{(2P_0)(2V_0)}{nR} = 4T_A$$

Heat added:

for diatomic gas $C_v = \frac{5}{2}R,\ C_p = \frac{7}{2}R$

$$Q_{AB} = nC_v(T_B - T_A) = n.\frac{5}{2}R.T_A$$

$$Q_{BC} = nC_p(T_C - T_B) = n.\frac{7}{2}R.(2T_A)$$

$$Q_{total} = nRT_A\left(\frac{5}{2} + 7\right) = nRT_A.\frac{19}{2}$$

Temperature change for whole process:

$\Delta T = T_C - T_A = 3T_A$

Average molar heat capacity C defined by $Q_{total} = nC\Delta T$:

$$nC(3T_A) = nRT_A.\frac{19}{2}$$

$$C = \frac{19}{6}R$$

32) Rate of cooling $\frac{\Delta\theta}{t} = \frac{A\varepsilon\sigma(T^4 - T_0^4)}{mc}$

$\Rightarrow t \propto \frac{m}{A}$ [$\because \Delta\theta$, t, σ, (T^4–T^{04}) are constant]

$\Rightarrow t \propto \frac{m}{A} \propto \frac{\text{Volume}}{\text{Area}} \propto \frac{a^3}{a^2}$

$\Rightarrow t \propto a \Rightarrow \frac{t_1}{t_2} = \frac{a_1}{a_2}$

$\Rightarrow \frac{100}{t_2} = \frac{1}{2} \Rightarrow t_2 = 200s$

33)

At steady state $\frac{dQ}{dt}$ = constant

$$\frac{KA(T_2 - \theta)}{x} = \frac{2KA(\theta - T_1)}{4x}$$

$$\Rightarrow \theta = \frac{2T_2 + T_1}{3}$$

So, $\frac{dQ}{dt} = \frac{KA\left(T_2 - \left(\frac{2T_1+T_1}{3}\right)\right)}{x}$

$\frac{dQ}{dt} = \frac{KA(T_2 - T_1)}{3x}$

$f = \frac{1}{3}$

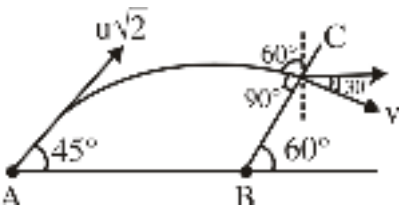
34) $\vec{a} - \vec{c} = \vec{c} - \vec{b}$

$2\vec{c} = \vec{a} + \vec{b}$

35) $y = \frac{a}{t^2} e^{-\left[\frac{t}{b} - \frac{x}{c}\right]}; \left[\frac{t}{b}\right] = \left[\frac{x}{c}\right] = M^0L^0T^0$

$\Rightarrow$ [b] = T & [C] = [L] & $\left[\frac{a}{t^2}\right] = y \Rightarrow [a] = \left[LT^2\right]$ & $e^{-\left(\frac{t}{b} - \frac{x}{L}\right)}$ is dimensionless

36)



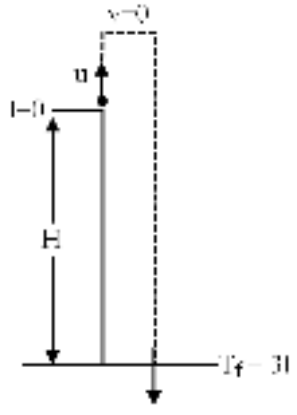
In horizontal direction, velocity remains same.

$v\cos 30° = u\sqrt{2}\cos 45°$

$v \times \frac{\sqrt{3}}{2} = u\sqrt{2} \times \frac{1}{\sqrt{2}}$

$v = \frac{2u}{\sqrt{3}}$

37) Time to reach to highest point = $\frac{u}{g} = \frac{5}{10} = \frac{1}{2}$ sec



$H = -(5) \times \left(\frac{3}{2}\right) + \frac{1}{2}(10)\left(\frac{3}{2}\right)^2$

$H = -\frac{15}{2} + \frac{90}{8}$

H = – 7.50 + 11.25

H = 3.75 m

38) $a_A = \frac{F}{2m}$ & $a_B = \frac{F}{4m}$

$\Rightarrow a_{AB} = \frac{F}{4m}$

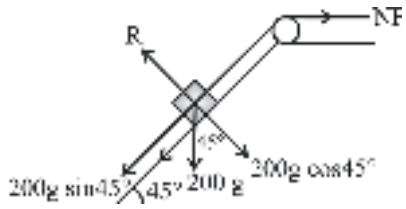




$\frac{1}{2}a_{AB} \times t^2 = L$

$\Rightarrow t = \sqrt{\frac{2L \times 4m}{F}}$

39)



Here, mass of the block, m = 200kg

coefficient of static friction, $\mu_s = 0.5 = \frac{1}{2}$

angle of incline plane, $\theta = 45°$

Maximum force that each man can apply, F = 500 N

Let N number of men are required for the block to just start moving up the plane NF = mgsinθ + f

= mgsinθ + μ_sR

= mgsinθ + μsmgcosθ = mg[sinθ + $\mu_s$cosθ]

$= 200 \times 10\left[\sin 45° + \frac{1}{2}\cos 45°\right]$

$= 200 \times 10\left[\frac{1}{\sqrt{2}} + \frac{1}{2\sqrt{2}}\right] = \frac{200 \times 10 \times 3}{2\sqrt{2}}$

$\therefore N = \frac{200 \times 10 \times 3}{2\sqrt{2} \times 500} = 5$.

40) Force produced by the engine

$F = \frac{P}{v} = \frac{30 \times 10^3}{30} = 10^3 N$

$\text{Acceleration} = \frac{\text{Forward force by engine-resistive force}}{\text{mass of car}}$

$= \frac{1000 - 750}{1250} = \frac{250}{1250} = \frac{1}{5} m/s^2$

41) COME : $K_1 + U_1 = K_2 + U_2$

$0 + mg(4R) = \frac{1}{2}mv^2 + mg(2R) \Rightarrow mv^2 = 4mgR$

Forces at position 2 :

N= $\frac{mv^2}{R}$ –mg = 4mg – mg = 3 mg

42)

disc B will climb on inclined plane only using its translational KE hance its maximum height must be less then that of A

43)

Maximum force in the string is $F_{max} = mg + \frac{mv^2}{\ell}$

$\because v = \sqrt{2g\ell}$, $F_{max} = 3\ mg$

$\Delta\ell = \frac{F\ell}{YA} = \frac{3mg\ell}{YA}$

$= \frac{3 \times 40 \times 5}{2 \times 10^{11} \times 2 \times 10^{-6}} = 1.5\,mm$

44) $\frac{T_2}{T_1} = \left(\frac{r_2}{r_1}\right)^{3/2} = \left(\frac{4R}{R}\right)^{3/2} = 8$

$\Rightarrow T_2 = 8T_1 = 8\sqrt{2}$ hr.

45)

Strain = $\frac{\Delta\ell}{\ell} = \left(\frac{2\pi R - 2\pi r}{2\pi r}\right)$



Strain = $\left(\frac{R-r}{r}\right)$

$Y = \frac{F/A}{\frac{\Delta\ell}{\ell}}$, $Y\frac{\Delta\ell}{\ell}A = F$

$F = AY\left(\frac{R-r}{r}\right)$

CHEMISTRY

46) For H-atom energy of electron depends on n-only.

47)

PQ_2	$\rightleftharpoons$	P +	2Q
600		0	0
600-P		P	2P

$P_T = 900$
$600 + 2P = 900$
$P = 150$
$(P.P)_Q = 2P = 300$

48) Negative sign represents decrease in concentration of reactants with time.

49) $K_1 = \frac{1}{R_1} \times G^*$ (KCl solution)

$K_2 = \frac{1}{R_2} \times G^*$ (K_2SO_4 solution)

$\Rightarrow \frac{K_1}{K_2} = \frac{R_2}{R_1}$

$\Rightarrow K_2 = \frac{3 \times 100}{300} = 1\,\Omega^{-1}\,m^{-1}$

$\Lambda_m = K_2 \times \frac{1000}{M \times 10^6} = 1 \times \frac{1000}{0.01 \times 10^6}$

$\Lambda_m = 10^{-1}\ \Omega^{-1}\,m^2 mol^{-1}$

50) **Question Explanation:** The question asks which solution of NH_4Cl shows the maximum degree of hydrolysis.

Concept: Degree of hydrolysis

Solution:
NH_4Cl :- SAWB type salt
For (SAWB) type salt hydrolysis -
degree of hydrolysis (h) $\propto \frac{1}{\sqrt{C}}$
C → min. , h → maximum.
0.01 M NH4Cl
0.1 M NH4Cl
0.001 M NH_4Cl → C is minimum, h is maximum

Final Answer: option (3)

51)

Urea → nonelectrolyte, i=1
$BaCl_2$ → dissociates into $Ba^{2+} + 2Cl^-$, i = 3
$AlCl_3$ → dissociates into $Al^{3+} + 3Cl^-$, i = 4
For equimolar solution, $\pi \propto i$.
Order : i_{AlCl_3} (4) > i_{BaCl_2} (3) > i_{urea} (1).

52) $P_S = P^o\ X_{solvent}$

$$\frac{684}{760} = X_{solvent}$$
$$0.9 = X_{solvent}$$

53)

$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

$$\frac{V_{C_3H_8}}{1} = \frac{V_{O_2}}{5}$$
$$V_{O_2} = 100(L)$$
$$V_{air} = 5 \times V_{O_2}$$
$$= 500(L)$$

54)

(A) 22.4 L CH_4 at STP

A. Volume = 1 mole → 1 mole of CH_4

B. CH_4 has 6 C + 4 H = 6 + 4 = 10 electrons (total electrons per molecule)
number of electrons = Total electrons x N_A
= 10 x N_A

C. Match → R

(B) 18 g H_2O(l)

A. Mass = 18 g → 1 mole → contains 1 × N_A molecules

B. Match → P (contains 8 × N_A neutrons, as H_2O has 8 neutrons in oxygen + 0 in H)

(C) 17 g NH_3(g)

A. Molar mass = 17 g/mol → 1 mole → contains 1 × N_A molecules → 4 atoms per molecule × N_A → total atoms = 4 × N_A

B. Match → Q

(D) 10 g HF (F = 19)

A. Molar mass = 1 + 19 = 20 g/mol

Mole = $\frac{10}{20}$ → 0.5 mole

B. Match → S

55) $IO_4^- \rightarrow I_2$ ⇒ v.f. of $IO_4^- = 7$
⇒ Eq. wt = M/7

56) $MnO_4^- + C_2O_4^{2-} \rightarrow Mn^{2+} + CO_2$
$n_f = 5$ $n_f = 2$
Number of equivalents of $KMnO_4$ = No. of equivalents of $C_2O_4^{2-}$
$1 \times 5 = x \times 2$
$x = \frac{5}{2}$

57)

$C_2H_{4(g)} + 3\ O_{2(g)} \rightarrow 2CO_{2(g)} + 2H_2O_{(\ell)}$ $\Delta H = -1420$ KJ
1420KJ → 1 mole
355 KJ → 0.25 mole
Volume at STP = 0.25 × 22.4 = 5.6L
OR
⇒ 1420 kJ heat evolved from 1 mole C_2H_4
⇒ 355 kJ heat evolved from $\frac{1}{1420} \times 355$ mole C_2H_4
$\frac{355}{1420} \times 22.4$ = 5.6 litre

58)

$w = nCv\ (T_2 - T_1)$
$-3 \times 1000 = 20\ (T_2 - 300)$
$T_2 - 300 = \frac{-3 \times 1000}{20} = -150$
$T_2 = 300 - 150 = 150$ K

59)

(i) Correct: Peptization is the conversion of a fresh precipitate into a colloid.
(ii) Incorrect: Lyophilic sols are typically organic (starch, gum). Inorganic sols are usually lyophobic.
(iii) Correct: Lyophilic means "solvent-loving," so particles are highly solvated.
(iv) Incorrect: Lyophobic sols are irreversible. Lyophilic sols are reversible.

60)

Phenolphthalein is a chemical compound used as an indicator in acid-base titrations. Its color changes depending on the pH of the solution.
• In acidic solutions, phenolphthalein is colourless.
• In basic (alkaline) solutions, it turns pink.

61) **Solution:**
(I) F The order of energy for a multi-electron atom like Mn is 3s < 3p < 3d [1]. The energies are equal only for a hydrogen atom or hydrogen-like ions.
(II) F The 5th period of the long-form periodic table contains 18 elements [1].
(III) T The ground state electronic configuration of Zr(Z = 40) is $[Kr]5s^2 4d^2$. For the Zr^{+2} ion, the configuration becomes $[Kr]4d^2$, which has $n = 2$ unpaired electrons. The magnetic moment is calculated using the spin-only formula $\mu s = \sqrt{n(n+2)}$ BM, resulting in $\mu s = \sqrt{2(2+2)} = \sqrt{8}$ BM.
(IV) F The general valence shell electronic configuration for *f*-block elements is $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$, where the *d*-orbital can have 0 or 1 electron, not a full 10 electrons [1, 2].
Final Answer: The correct option is (3)

62)

Ionic size :- $H^- > F^-$
* Alkali metal > Alkaline earth metal ⇒ Electro positive character.
* $\boxed{E.N. \propto \frac{+ve}{-ve}}$
* For ions of same element.

63) M.Pt → $NH_3 > SbH_3 > AsH_3 > PH_3$

64)

B.A. = NH_4^+ (sp³, '0' l.p.) > NH_2^- (sp³, 2 l.p.)
* $OF_2 < OCl_2$ ⇒ Size of side atom ↑ : B.A. ↑
* NO_2^+ (sp) > NO_2^- (sp^2) B.A.
* BF_3 (sp^2) = BCl_3 (sp^2) ⇒ B.A.

65) * Molecular orbitals should have maximum overlap
* Symmetry about the molecular axis should be similar

66) (a) SF_4 – sp^3d hybridisation
(b) $[Ni(NH_3)_6]^{+2}$ – sp^3d^2 hybridisation
(c) NO_2^+ – sp hybridisation
(d) MnO_4^- – sp^3 hybridisation

67)

*$M(AB)_3$ ⇒ Show both G.I. & optical isomer.
*$Ni(CO)_4$ ⇒ sp^3 no. of 109º28' B.A.. = 6
In metal carbonyl ; metal-carbon bond
passess both σ & π – character.

68) Solution: Cr^{3+} having $3d^3$ configuration always have 3 unparied electrons with strong field as well as weak field ligands.

69)

Ziegler catalyst → Titanium
Blood pigment → Iron
Wilkinson catalyst → Rhodium
Vitamin B_{12} → Cobalt

70) (A) Zn, Cd, Hg exhibit lowest enthalpy of atomization in respective transition series.
(C) Compounds of Zn, Cd and Hg are diamagnetic in nature.

71) The +3 oxidation state of lanthanide is most stable and therefore lanthanide in +4 oxidation state has strong tendency to gain e– and converted into +3 and therefore act as strong oxidizing agent.
eg. Ce^{+4}
And therefore CeO_2 is used to oxidized alcohol aldehyde and ketones.
Lanthanide in +2 oxidation state has strong tendency to loss e– and converted into +3 oxidation state therefore act as strong reducing agent.
 $EuSO_4$ act as strong reducing agent.

72) H_2S + oxidizing agent → S (yellow turbidity)

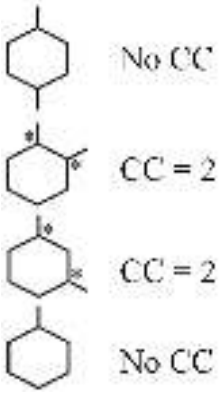
73) Hexagonal layered structure of graphite having mobile π-electron

74) In trivalent state most of the compounds being covalent are hydrolysed in water. Trichlorides on hydrolysis in water form tetrahedral $[M(OH)_4]^-$ species, the hybridisation state of element M is sp^3.
In case of aluminium, acidified aqueous solution forms octahedral $[Al(H_2O)_6]^{3+}$ ion.

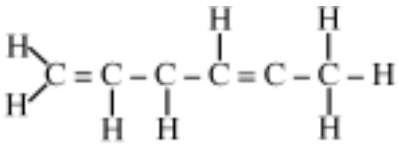
75) (P) $Br^- \xrightarrow{conc\ H_2SO_4} Br_2$ Brown
(Q) $S^{-2} \xrightarrow[Dil]{HCl} H_2S$ colorless rotten egg smell
(R) $NO_2^- \xrightarrow[HCl]{Dil} NO_2$ brown paramagnetic gas

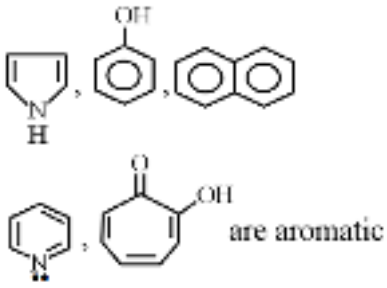
(S) $NO_3^- \xrightarrow{conc.H_2SO_4}$ brown paramagnetic gas

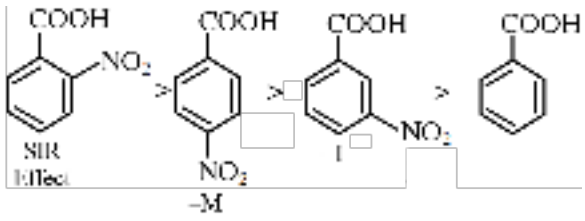
76) NCERT-XI Pg # 363 Part # II

77) 

Total chiral center = 4

78)  σ bonds = 13
π bonds = 3

79) 

80) 

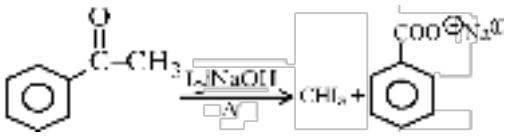
81) A–  ; B = n-Hexane ; C= Benzene

82)

$CH_3–CH_2–OH \xrightarrow[413K]{H_2SO_4} CH_3–CH_2–O–CH_2–CH_3$

83)

Iodoform reaction



84)

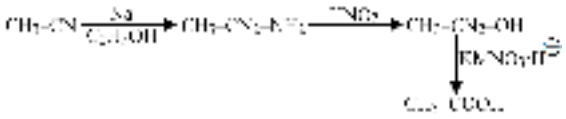
$SN^2 \propto \frac{1}{\text{Steric hindrance}}$

SN' ∝ Stability of carbocation

85)

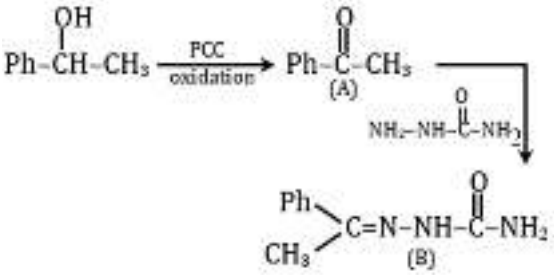
Reactivity of nucleophilic substitution for acid derivatives.
Acyl chloride > Acid anhydride > Ester > Amide

86)



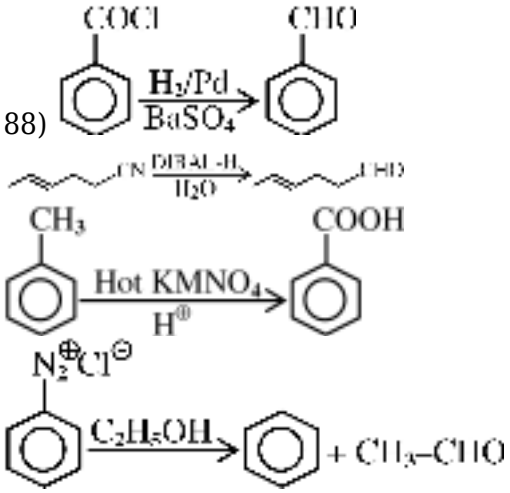
87)

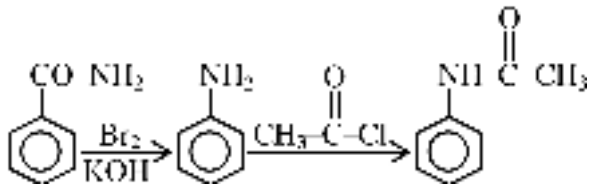
Solution :-



Final answer :- (2)

88)



89) 

90) NCERT Pg. # 418

BIOLOGY

91)

NCERT XIth Page No.29,30

92)

NCERT XIth Page No.4

93) **Explain Question: Question asks whether both statements about Mycoplasma are true or false.**
Concept: Mycoplasma
Solution: I. Mycoplasma are the smallest living cells and can survive without oxygen (facultative anaerobes)-CORRECT
II. Many Mycoplasma species are pathogenic to animals and plants-CORRECT
Final answer: 4

94)

NCERT XI, Pg. # 12,13 & 14

95) NCERT XI, Pg # 24

96)

NCERT XIth Page No.15

97) NCERT Pg # 26, 28, 30, 32

98)

NCERT XI, Pg # 95

99)

NCERT - XI Pg # 95, 96

100)

NCERT-XI, Pg. # 95

101) **Explaining the Question –** You are given a stage of cell division (image-based) and asked to select the correctly matched statement from the following options.
Concept – This Question is based on Anaphase in mitosis
Solution – Anaphase – Splitting of centromere and movement of daughter chromosomes to opposite polesThis is a defining event of anaphase in mitosis Centromeres split, and the now-separated chromatids move to opposite poles.
Final Answer - 3

102)

NCERT (XIth) Pg. # 165, 168

103)

NCERT XI, Page No.145

104)

Generated by Allie
Problem Statement: The question is asking which process occurs first during photosynthesis from the given options.
Underlying Concept: Photosynthesis involves multiple stages starting with the absorption of light energy by chlorophyll molecules. The initial reaction is the excitation of chlorophyll molecules caused by light absorption, which triggers subsequent steps like photolysis, ATP formation, and carbon fixation.
Tips and Tricks: Remember that the very first step is always the absorption of light by chlorophyll, which excites electrons to higher energy states.
Common Mistakes: Confusing photolysis (breaking water) as the first step, or assuming ATP formation happens before excitation.
Why Other Options Are Incorrect?: Photolysis of water occurs after the excitation of chlorophyll to replace lost electrons. ATP formation requires energy from excited electrons and thus occurs later. Fixation of CO_{2} happens in the Calvin cycle, which is after the light reactions.

105) NCERT Pg. # 228, 231

106)

First reaction in photosynthesis is excitation of chlorophyll molecule

107) NCERT-XI, Pg # 248

108)

NCERT Pg.# 247,248

109)

NCERT XII, Pg. # 12

110)

Question Explanation: Identifying the correct function for a labeled cell/structure in the embryo sac diagram.
Concept: Embryo Sac Structure and Function
Solution: The diagram shows the mature embryo sac.
A (Filiform apparatus): Guides pollen tube entry. (Incorrectly labeled D)
B (Synergids): Guide pollen tube entry. (Incorrectly labeled B)
C (Polar nuclei): Fuse with male gamete to form PEN. (Incorrectly labeled C)
D (Egg cell): Fuses with male gamete to form zygote. (Incorrectly labeled D)
Let's check the options again based on the provided labels:
A: Filiform apparatus.
B: Synergids.
C: Polar nuclei.
D: Egg cell.
E: Antipodal cells.
Option 2 is the most logically derived answer, stating the function of the Synergid (B) (Incorrectly referencing entry of pollen tube but indicating a function of the complex):
Final Answer: B = Entry of pollen tube in embryosac

111)

NCERT XIIth Page No.16

112) **Question Explanation :** The question is asking: “Are the given assertion and reason true, and does the reason correctly explain the assertion?”
Concept: Predation
Solution:
Assertion (A): Predators keep prey population under control. True
Reason (R): In the absence of predators, prey could achieve high population density causing ecosystem instability. True
Does the reason correctly explain the assertion? Yes — predators control prey populations exactly to prevent ecosystem instability caused by overpopulation.
Therefore, the correct option is: 1. Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
Final answer: The correct option is: 1. Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

113)

NCERT XIIth Page No.210

114)

NCERT XIIth Page No.212

115) Question Explanation: Matching the rationales for biodiversity conservation with their correct categories.
Concept: Reasons for Biodiversity Conservation
Solution: The reasons for conserving biodiversity are grouped into three categories:

A. Narrowly utilitarian: Direct economic benefits like food, firewood, and medicines. → (ii) Direct economic benefits.
B. Broadly utilitarian: Indirect, major ecosystem services like oxygen production, pollination, and aesthetic pleasure. → (i) Ecosystem services (oxygen, aesthetic value).
C. Ethical: Based on moral duty to protect every species, acknowledging their intrinsic value. → (iii) Intrinsic value (moral duty).

Final Answer: Option -2

116)

Column -I		**Column -II**	
(A)	Dodo	**(iii)**	Mauritius
(B)	Quagga	**(i)**	Africa
(C)	Thylacine	**(iv)**	Australia
(D)	Steller seacow	**(v)**	Russia
(E)	Tiger sub-species	**(ii)**	Bali, Javan, Caspian

117) NCERT XII, Pg. # 219

118) NCERT (XIth) Pg # 144

119)

NCERT XIth Page No.117

120) NCERT Page No. 159, (9.12.6)

121) **Solution-**
False statement is-Leaf is not a transpiratory organ.
The correct answer is -(4)

122)

NCERT XI, Pg. 66

123)

NCERT XIth Page No.65

124) Correct Answer: (3)

125)

Explain Question : The question asks to evaluate three statements about the sex determination and chromosome complement in grasshoppers and determine which are true or false.
Concept : This question is based on sex determination
Explanation :
(a) Total number of chromosomes are same in both ♂ and ♀ - Incorrect
Males have one less chromosome (no Y or second X)
So, total chromosome number differs
Female: 2A + XX
Male: 2A + XO
(c) Female heterogamety is present - Incorrect
In grasshoppers, male is heterogametic (XO)
Female is homogametic (XX)
Female heterogamety (ZW system) is seen in birds, not grasshoppers
Final answer : 3

126) NCERT Pg. # 75(E), 83(H)

127) **Asking about :**
this question is asking about the nature of Gregor Mendel's experiments and the type of traits he studied :
Concept :
The question involves understanding the difference between :
Qualitive traits : Traits that are categorical and distinct with no intermediate forms. Examples include flower color (red, white, purple) or seed shape (round, wrinkled).
Quantitative traits : Traits that show continuous variation and can be measured numerically. Examples include height, weight, or yield.
Solution :
Mendel's work forcused on **qualitative traits** in pea plants He meticulously analyzed traits like flower color, seed shape, and pod color, which had distinct categories. He did not study traits that showed continuous variation.
Answer : There fore, the true statement for Mendel is :
1. Qualitative study of qualitative characters

128)

NCERT-XII, Pg. # 57

129)

Explain Question: The question asks to select the false statement for independent assortment.

Concept : Dihybrid Cross

Solution: The incorrect statement is: Due to independent assortment a dihybrid produces 4 types of gametes in the ratio of 9 : 3 : 3 : 1. (The ratio represents Zygotes and not gametes)

So, the correct answer is: 3.

Final answer: 3.

130)

Question Explanation: This question asks to match different types of RNA (rRNA, tRNA, snRNA, mRNA) with their correct functions.

Concept: Split Gene

Solution:
A–i → rRNA – Peptide bond formation
B–ii → tRNA – Codon identification
C–iii → snRNA – Splicing
D–iv → mRNA – Genetic information
So, the correct answer is : A-i, B-ii, C-iii, D-iv

Final Answer: A-i, B-ii, C-iii, D-iv

131) NCERT, Pg. # 93/102

132) NCERT Pg. # 105-106

133) NCERT XII, Pg. # 99

134) NCERT (XII) Pg # 96

135)

NCERT-XII Pg#105

136) NCERT Pg. # 45

137) NCERT Pg. # 42

138) NCERT Pg. # 41, 42, 45, 44

139) NCERT Pg. # 48

140)

NCERT Pg. # 48

141) NCERT Pg.# Module/158.

142) NCERT Pg#82,83

143) NCERT Page No. 202, 203

144) NCERT Pg#200,201,202

145) NCERT Pg#201

146) NCERT XI Page No. # 194, 195

147) NCERT, Pg. # 189

148) NCERT, Pg. # 187

149) NCERT Pg. No. # 189, 190

150)

NCERT Pg. # 209

151) NCERT-XI, Pg. # 205

152) NCERT, Pg. # 207

153) NCERT Pg # 317,318,333

154) NCERT XI Page No. # 234, 235, 243, 245

155) NCERT (XII) ; Page No. # 230

156) NCERT Pg.#244

157) NCERT Pg.#245, 246

158) NCERT, Pg # 241, 242

159) NCERT Pg.#246

160) NCERT Pg.#228

161) NCERT XI Pg.#222

162) NCERT (XII) ; Page No. # 226

163)

NCERT Pg. # 27

164) NCERT(XII) Chapter # 2

165) NCERT XII Pg. No. # 37, 38

166) NCERT(XII) Page#27/29(H) Fig:2.1(b)

167)

NCERT Pg. # 44

168) NCERT Pg. No. # 137-138

169) NCERT XII Pg. No. # 136

170) NCERT Pg. 137, 138

171) Module

172) NCERT XII Bio P-112

173) NCERT(XII) Page#44/50(H) Para:3.2

174) NCERT XII Pg. No. # 124

175) NCERT-XII, Pg, # 118

176) NCERT Pg. 183

177)

NCERT Pg. # 166

178)

NCERT-XII, Pg. # 184

179) NCERT-XII, Pg. # 179

180) NCERT XII, Pg. # 153